

Review

UWB-Based Real-Time Indoor Positioning Systems: A Comprehensive Review

Mohammed Faeik Ruzaij Al-Okby ^{1,2} , Steffen Junginger ^{3,*}, Thomas Roddelkopf ³ and Kerstin Thurow ¹ 

¹ Center for Life Science Automation (Celisca), University of Rostock, 18119 Rostock, Germany; mohammed.al-okby@celisca.de or mohammed.al_okby@atu.edu.iq (M.F.R.A.-O.); kerstin.thurow@celisca.de (K.T.)

² Technical Institute of Babylon, Al-Furat Al-Awsat Technical University (ATU), Kufa 54003, Iraq

³ Institute of Automation, University of Rostock, 18119 Rostock, Germany; thomas.roddekopf@uni-rostock.de

* Correspondence: steffen.junginger@uni-rostock.de; Tel.: +49-(381)-498-7808

Abstract: Currently, the process of tracking moving objects and determining their indoor location is considered to be one of the most attractive applications that have begun to see widespread use, especially after the adoption of this technology in some smartphone applications. The great developments in electronics and communications systems have provided the basis for tracking and location systems inside buildings, so-called indoor positioning systems (IPs). The ultra-wideband (UWB) technology is one of the important emerging solutions for IPs. This radio communications technology provides important characteristics that distinguish it from other solutions, such as secure and robust communications, wide bandwidth, high data rate, and low transmission power. In this paper, we review the implementation of the most important real-time indoor positioning and tracking systems that use ultra-wideband technology for tracking and localizing moving objects. This paper reviews the newest in-market UWB modules and solutions, discussing several types of algorithms that are used by the real-time UWB-based systems to determine the location with high accuracy, along with a detailed comparison that saves the reader a lot of time and effort in choosing the appropriate UWB-module/method/algorithm for real-time implementation.

Keywords: object tracking; ultra wideband (UWB); indoor positioning systems (IPs); Internet of Things (IoT); real-time location system (RTLS); localization; mobile robot



Citation: Al-Okby, M.F.R.; Junginger, S.; Roddelkopf, T.; Thurow, K. UWB-Based Real-Time Indoor Positioning Systems: A Comprehensive Review. *Appl. Sci.* **2024**, *14*, 11005. <https://doi.org/10.3390/app142311005>

Academic Editors: Višnja Križanović, Krešimir Grgić and Drago Žagar

Received: 14 October 2024

Revised: 18 November 2024

Accepted: 24 November 2024

Published: 26 November 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Currently, the use of autonomous navigation devices and equipment is common in the industrial and scientific environment. The use of mobile robots is no longer limited to large factories and universities. Their use has become very widespread in all aspects of life, such as hospitals, shopping malls, airports, restaurants, and even homes. Autonomous navigation systems for mobile robots and moving devices need to provide constantly updated location information to optimally ensure accurate movement and a safe approach to the target destination. Furthermore, the movement of robots or self-driving equipment within these facilities may encounter some obstacles, leading to collisions that may result in damage of varying severity. Therefore, the need arises to use accurate tracking systems for their movement inside buildings and related facilities. These systems are commonly called indoor positioning systems (IPs), real-time location systems (RTLSs), and real-time tracking systems (RTTs).

Positioning can be defined as the process of using technology to monitor and determine the exact location of objects, robots, or people while stationary or moving. The first generations of positioning systems were able to efficiently track objects in outdoor environments based on the use of satellite technologies to provide Global Navigation Satellite System (GNSS) services such as GPS, GLONASS, Baidu, Galileo, and others.

The concept of indoor positioning systems, which represent the second type of positioning systems, has evolved with the development of devices and equipment operating with Internet of Things technologies. IoT solutions have provided the possibility of carrying out these complex tasks with high efficiency and reliability. The idea of developing indoor positioning systems emerged due to the inability of the Global Navigation Satellite System to track and determine the location of objects inside buildings. This results from the attenuation of signals sent and received, due to the walls of buildings, as well as the large margin of error of several meters. This makes the use of these techniques impractical due to low accuracy, as some applications inside buildings require high accuracy and an error rate that does not exceed several centimeters [1].

The attractiveness of the topic of developing indoor positioning systems has prompted many researchers and companies to develop different feasible solutions to achieve this goal. The proposed solutions rely on the use of available IoT-based technologies such as Bluetooth Low Energy [2], Wi-Fi [3], ultra-wideband technology [4], vision-based techniques [5], acoustic waves [6], radio frequency identification (RFID) [7], and others. Choosing the appropriate IPS for determining object locations in indoor environments depends primarily on the nature of the place to be covered, in terms of area, and the presence of obstacles such as walls and furniture. The nature of an open space such as a warehouse or stadium differs from that of inhabited buildings such as laboratories and residences, which contain narrow corridors and large amounts of furniture (obstacles). The choice also depends on the type of application, the required accuracy, the speed of updating data, and the cost allocated for implementation.

In this work, a deep and focused search was conducted in the literature to review research articles, conference papers, book chapters, and studies carried out practically for finding the positioning and tracking the motion of objects using ultra-wideband technology, as this is considered to be one of the best solutions currently available. Unlike other radio frequency waves (RF), Wi-Fi, and Bluetooth, the ultra-wideband waves are characterized by enabling accurate measurement of the time of flight between its units and, thus, the possibility of accurate measurement of distance, direction, and location determination with a relatively small margin of error compared to other RF waves. Ultra-wideband waves have a much larger bandwidth than narrowband waves. This results in ultra-wideband waves having a very short wavelength, enabling the temporal accuracy to be very high, which is reflected in high accuracy in measuring time of flight and also contributes to improving the resistance to multipath interferences and fading. Furthermore, the large bandwidth and low propagation time make the use of ultra-wideband waves ideal for real-time systems, especially those that require a quicker response time, as they are about 50 times faster than GPS systems and hundreds of times faster than traditional Bluetooth systems. These characteristics make ultra-wideband the ideal choice in automation applications, as well as applications that deal with fast movement of objects to be tracked and high-speed applications [8]. Due to the large frequency range where UWB transmissions are allowed (3.1–10.6 GHz), the transmission power was limited to -41.3 dBm to avoid and reduce interference with waves of other technologies operating within the same frequency range. Figure 1 summarizes the main characteristics of UWB and other RF technology frequencies and power spectral densities.

Several keywords were used to import the related research works—such as “UWB, Positioning, Localization, Tracking, IPS, RTLS, RTTS”—from several databases, including IEEE, Elsevier (Scopus and ScienceDirect), Clarivate Web of Science (WoS), PubMed, and the ACM Library, during the past five years between January 2019 and June 2024. More than 270 research papers were retrieved from the aforementioned sources to examine their suitability for inclusion in this study. The inclusion/exclusion criteria were based on several factors related to the nature of the system used and the modernity of the work. The importance of the present work lies in providing a comprehensive view of indoor positioning systems operating with ultra-wideband technology in real time, and enabling researchers and engineers interested in this type of system to identify the available

commercial units and their technical specifications, as well as the methods, techniques, and algorithms that enable the reader to reduce the effort and time expended in developing the relevant systems.

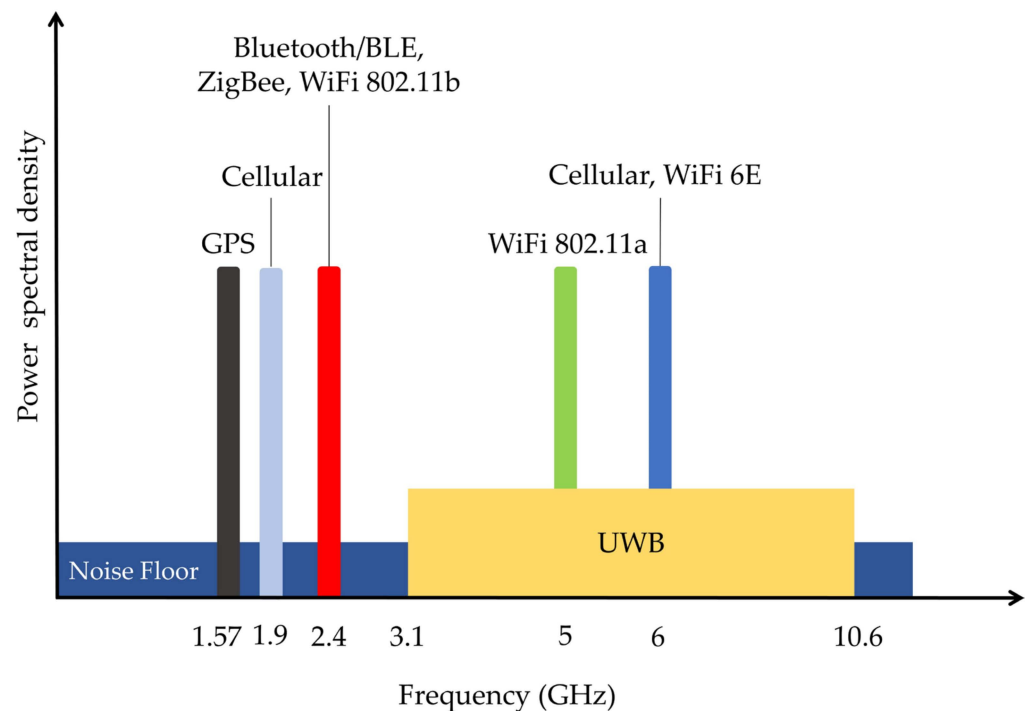


Figure 1. UWB and other RF technologies' frequencies and power spectral densities.

2. Materials and Methods

In this work, the focus was on real-time indoor positioning systems based on ultra-wideband technology, because of the important advantages that they offer, which make them one of the most effective, robust, and practical solutions for indoor positioning, localization, and tracking tasks for mobile objects. Furthermore, this paper concentrates on the available UWB chips, modules, and solutions that can be used in the real-time implementation of object tracking, positioning, and localization applications.

2.1. Search Strategy

The present work includes the available research references for the past 5 years, from January 2019 until August 2024. Several keywords were used to import the most relevant articles, conference papers, book chapters, and studies, such as object tracking, localization, positioning, indoor object tracking, indoor localization, indoor positioning, indoor positioning system, real-time tracking system, real-time localization system, ultra-wideband, UWB tracking, multi-tag positioning, time of arrival (ToA), time of departure (ToD), time difference of arrival (TDoA), angle of arrival (AoA), phase difference of arrival (PDoA), two-way ranging (TWR), multilateration, trilateration, and triangulation. The search was initially started on the IEEE Explore and ScienceDirect databases and then extended to the other listed databases. More than 600 related papers were imported from IEEE, Elsevier (Scopus and ScienceDirect), WoS, PubMed, and the ACM Library for the mentioned period for eligibility. A group of the most relevant papers was selected to be discussed in the present work. We focused on several points that helped with the inclusion and exclusion criteria. The systematic review was carried out following the PRISMA protocol.

2.2. Inclusion and Exclusion Criteria

The term “real-time” refers to the localization, tracking, and positioning systems that have been practically implemented, tested, and can be used in real-time implementation. Furthermore, limited the inclusion of the selected research to systems for indoor environment applications. Since many technologies can be used to build indoor positioning systems, the focus was on scientific research and systems that operate with ultra-wideband technology, especially those used for mobile object tracking applications such as mobile robots and unmanned aerial vehicles (UAVs). Accordingly, the inclusion and exclusion criteria were as follows:

- Inclusion criteria
 - Real-time systems, positioning algorithms, and filters for a real-time system.
 - Systems for indoor application.
 - Can be used for mobile object tracking (robots, UAVs, etc.).
 - Complete IPS/RTLS system including hardware, software, and monitoring server (object coordinates, tracking, and positioning can be monitored).
- Exclusion criteria
 - Simulation, theoretical research, and algorithms; UWB manufacturing research.
 - Systems for outdoor environments.
 - Systems cannot be used for mobile object applications.
 - Partial work (incomplete system; cannot implement real-time tracking and monitoring; work for solving a partial issue).
 - Works that do not include any information about the UWB components used.

Based on the inclusion criteria, 147 papers were included that met the aims of the proposed study. Figure 2 illustrates the flowchart of the system selection process.

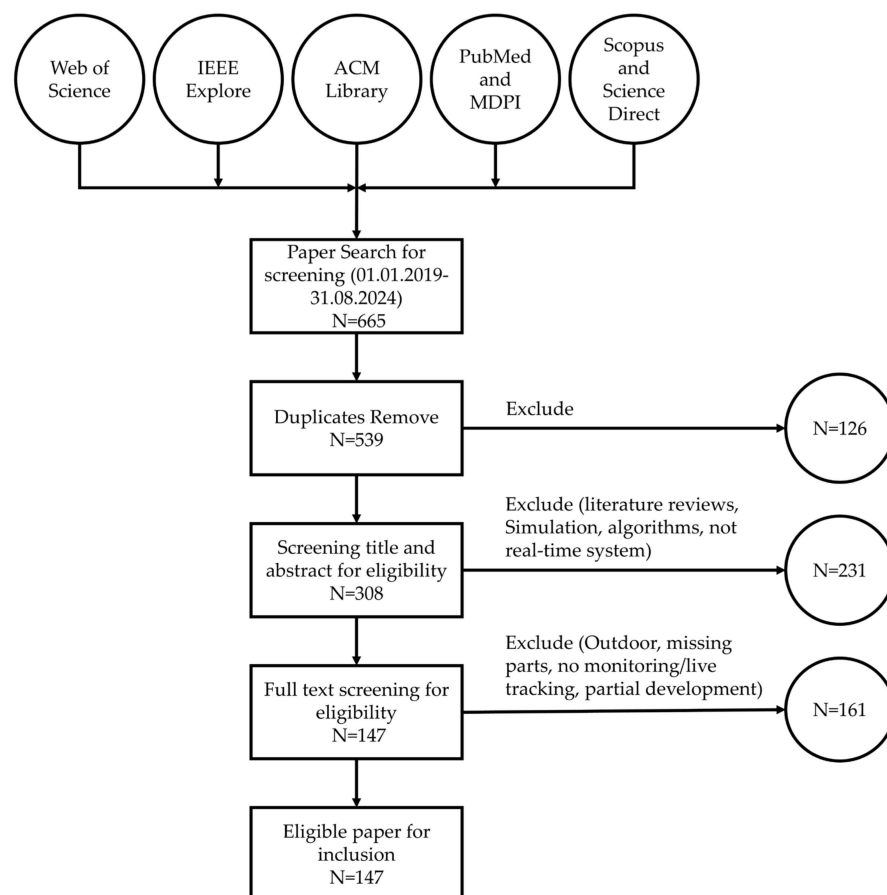


Figure 2. Flowchart of the selection process.

3. UWB Positioning System

Ultra-wideband systems have emerged as a promising technology in the field of position monitoring, especially for moving objects within a closed indoor environment. The quality of the pulses issued by ultra-wideband transmitters is characterized by high temporal accuracy, which makes it possible to measure the transmission and reception time (time of flight) easily, enabling high precision in determining the location. The accuracy can vary from a few decimeters to centimeters in optimal conditions [9]. Most position detection systems usually consist of two main components, which may be identical in terms of design and physical components but different in terms of the method of use within the system: the anchor, and the tag.

In most of the current systems, anchors are used as receivers for the signal issued by the tag, and to determine the distance between the anchor and tags and then forward the measurement to the IPS controller. Numerous anchors are used, depending on the area of the place to be covered. Currently, the coverage of a single anchor operating with ultra-wideband technology has reached approximately 100 m in line-of-sight conditions (LoS: the state of having a direct connection between the sender and the receiver without any obstacles). This contributes to reducing the cost of the infrastructure used, which represents one of the factors that limit the use of these systems. Anchors are installed in a specific way and with dimensions that depend on the type of method or algorithm used to determine the position.

3.1. UWB Positioning Techniques and Algorithms

Several techniques can be used with ultra-wideband waves to determine the indoor position of movable objects. In general, the UWB location tracking of the unknown movable tag can be determined using one of the following basic techniques: triangulation, or trilateration [10]. These techniques have been derived from ancient mathematical calculations to determine the locations of things.

3.1.1. Triangulation Technique

The triangulation technique is based on calculating angles. The location of a moving tag can be identified if the location of two fixed anchors is known, by estimating the angles between the movable tag and the antennas of the fixed anchors and intersecting the extension of the moving tag with the known line between the two fixed anchors, which enables the accurate determination of the location of the moving tag (see Figure 3).

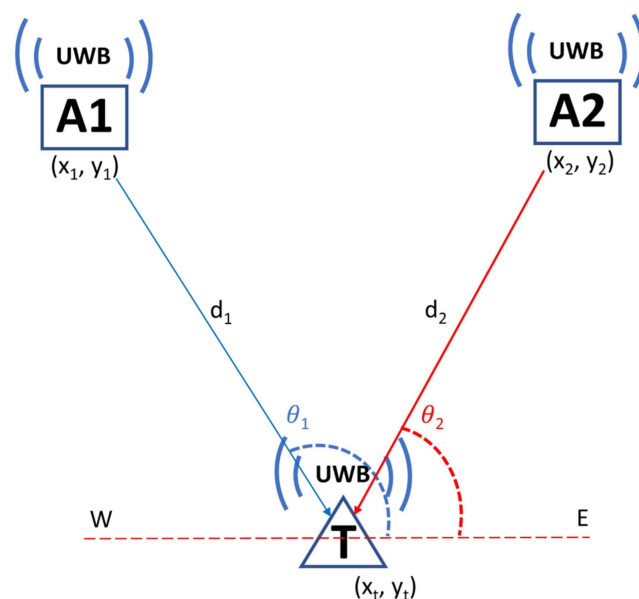


Figure 3. Triangulation IPS positioning technique.

In Figure 3, the coordinates of anchor1 and anchor2 are (x_1, y_1) and (x_2, y_2) , respectively, and the coordinates of the movable tag are (x_t, y_t) . Two angles were observed between the anchors, the movable tag, and the west–east line:

$\theta_1 = A1\ T\ E$ (the angle between anchor1 (A1), tag (T), and E horizontal line);

$\theta_2 = A2\ T\ E$ (the angle between anchor2 (A2), tag (T), and E horizontal line).

The slope of the line A1 T can be calculated using the following formula:

$$\tan\theta_1 = \frac{y_t - y_1}{x_t - x_1} \quad (1)$$

We can repeat the same procedure for the line A2 T:

$$\tan\theta_2 = \frac{y_t - y_2}{x_t - x_2} \quad (2)$$

By solving Equations (1) and (2), we get

$$x_t = \frac{(y_1 - y_2) + x_2 \tan\theta_2 - x_1 \tan\theta_1}{\tan\theta_2 - \tan\theta_1} \quad (3)$$

and

$$y_t = \frac{[y_1 \tan\theta_2 - y_2 \tan\theta_1] - [(x_1 - x_2) \tan\theta_2 \tan\theta_1]}{\tan\theta_2 - \tan\theta_1} \quad (4)$$

3.1.2. Trilateration Technique

The trilateration technique depends on calculating the distances between the moving tag and the fixed anchors. The distance is calculated by multiplying the time of wave propagation (ToF) by the speed of the UWB wave, which is approximately $(3 \times 10^8\ \text{m/s})$. If the coordinates of the used anchors (which are predefined by the user) and the distances between the movable objects and anchors can be provided by the used technology, then we can find and track the object location using trilateration techniques.

The distance d between any two points can be calculated with the known coordinates (x, y) of the two points; for example, the distance between (x_1, y_1) and (x_2, y_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (5)$$

If the tag coordinates are (x_t, y_t) and the anchor coordinates are (x_a, y_a) , we can rewrite the equation as follows:

$$\begin{aligned} d_a^2 &= (x_a - x_t)^2 + (y_a - y_t)^2 \\ d_a^2 &= x_a^2 + x_t^2 - 2x_a x_t + y_a^2 + y_t^2 - 2y_a y_t \end{aligned} \quad (6)$$

when a point (r) is considered as a reference point for the IPS and used to fix an anchor with the coordinates (x_r, y_r) , and the remaining anchors have the coordinates (x_i, y_i) , where i represents the anchor number. Subtracting the d_r from d_i results in:

$$d_r^2 - d_i^2 + x_i^2 + y_i^2 - x_r^2 - y_r^2 = 2(x_i - x_r)x_t + 2(y_i - y_r)y_t \quad (7)$$

With $r = 1$ and $a = 2, 3, \dots$ (i represents all of the used anchors), we obtain

$$\begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) \\ 2(x_3 - x_1) & 2(y_3 - y_1) \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} d_1^2 - d_2^2 + x_2^2 + y_2^2 - x_1^2 - y_1^2 \\ d_1^2 - d_3^2 + x_3^2 + y_3^2 - x_1^2 - y_1^2 \end{bmatrix} \quad (8)$$

By representing Equation (8) by the linear system equation $(Ax = b)$, x will be the coordinate of the movable tag (x_t, y_t) , which can be processed and monitored using a proper positioning algorithm and monitoring program. Figure 4 explains the structure of the trilateration technique for a three-anchor IPS.

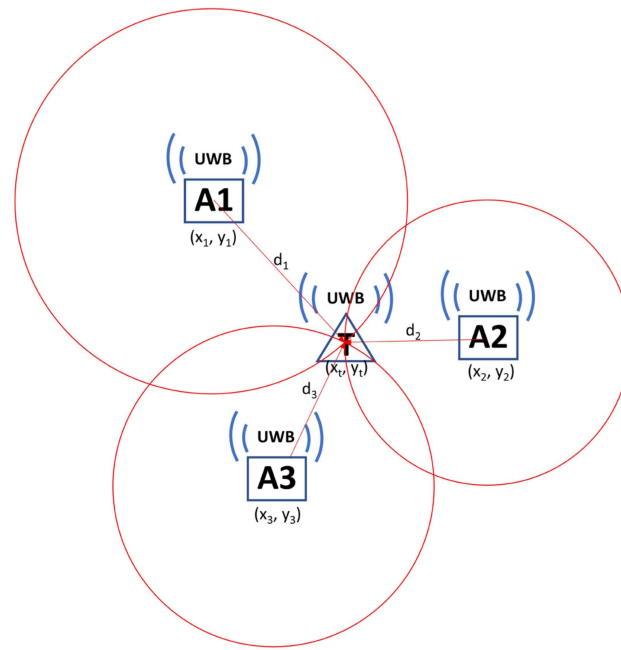


Figure 4. Trilateration IPS positioning technique.

The great industrial development and the effective introduction of robots into production processes has coincided with the need for three-dimensional (3D) positioning systems. These systems differ from the common two-dimensional systems in the need to calculate the height (h) of the moving tag in addition to the (x, y) coordinates (see Figure 5). Some 3D positioning systems require an additional anchor fixed at a different height than the rest of the IPS anchors to measure the height of the moving tags. From Equation (8), if we suppose that the number of the anchors used in the system is (i), we can calculate the 3D position coordinates of the movable tag (x_t, y_t, h_t) as follows:

$$\begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) & 2(h_2 - h_1) \\ 2(x_3 - x_1) & 2(y_3 - y_1) & 2(h_3 - h_1) \\ \vdots & \vdots & \vdots \\ 2(x_i - x_1) & 2(y_i - y_1) & 2(h_i - h_1) \end{bmatrix} \begin{bmatrix} x_t \\ y_t \\ h_t \end{bmatrix} = \begin{bmatrix} d_1^2 - d_2^2 + x_2^2 + y_2^2 + h_2^2 - x_1^2 - y_1^2 - h_1^2 \\ d_1^2 - d_3^2 + x_3^2 + y_3^2 + h_3^2 - x_1^2 - y_1^2 - h_1^2 \\ \vdots \\ d_1^2 - d_i^2 + x_i^2 + y_i^2 + h_i^2 - x_1^2 - y_1^2 - h_1^2 \end{bmatrix} \quad (9)$$

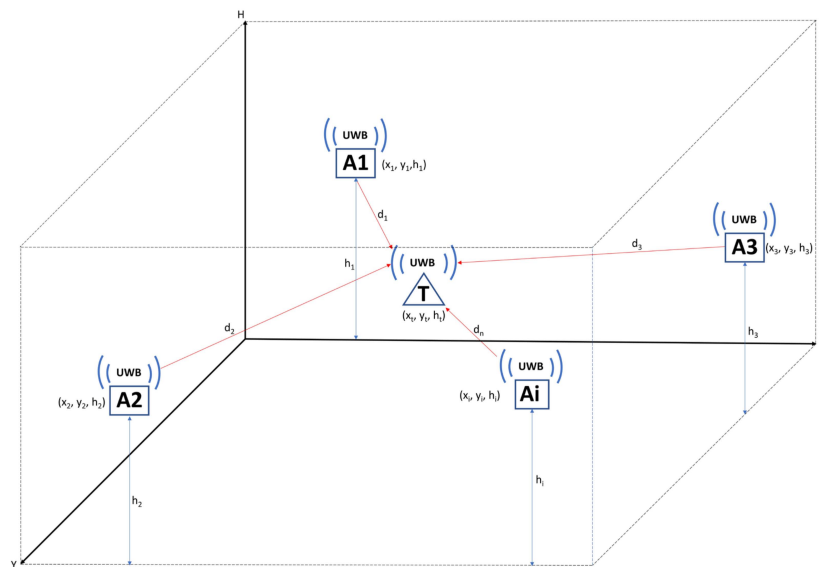


Figure 5. Multilateration positioning technique in a 3D environment.

3.2. UWB IPS Topologies

The following subsections discuss in detail selected main topologies used for UWB IPSs.

3.2.1. Two-Way Ranging (TWR)

Two-way ranging is a basic and effective method for calculating indoor objects' position. The distance between an anchor and a tag is calculated by measuring the time of flight (ToF) of ultra-wideband waves between them, where the time of flight of the waves between the sender and the recipient is multiplied by the speed of light (the speed of propagation of radio waves). There are several methods to implement TWR, including single-sided TWR (SS-TWR), symmetrical double-sided TWR (SDS-TWR), alternative double-sided TWR (AltDS-TWR), and asymmetrical double-sided TWR (ADS-TWR). In a two-coordinate medium, at least three anchors are required to calculate the positioning of the tag, where each tag–anchor distance is calculated separately and represents the radius of a tag–anchor circuit. The tag position can be accurately calculated by intersecting the three tag–anchor circles, where the intersecting point represents the object's expected position [11–13]. Figure 6 explains the UWB signal propagation through the distance calculation phase in SDS-TWR. To measure the distance between an anchor and a tag, three messages need to be exchanged. The tag starts the TWR by sending the first poll message request to the selected anchor, which takes the time of flight t_1 to reach the anchor. The anchor requires the time Td_1 (delay time) to record and process the request and replay the response to the tag. The total time from the tag request until the tag receives the anchor response is the round time $Tr_1 = t_1 + Td_1 + t_2 = 2ToF + Td_1$. The tag records the time Tr_1 and composes and sends the final set message, including the final data packet, which includes all of the required time data to calculate the distance by the anchor. The second-round time Tr_2 starts from the anchor response until the anchor receives the final packet $Tr_2 = t_2 + Td_2 + t_3 = 2ToF + Td_2$. From Figure 6, we can calculate the UWB signal's ToF using the following equation [14]:

$$ToF = \frac{((Tr_1 + Tr_2) - (Td_1 + Td_2))}{4} \quad (10)$$

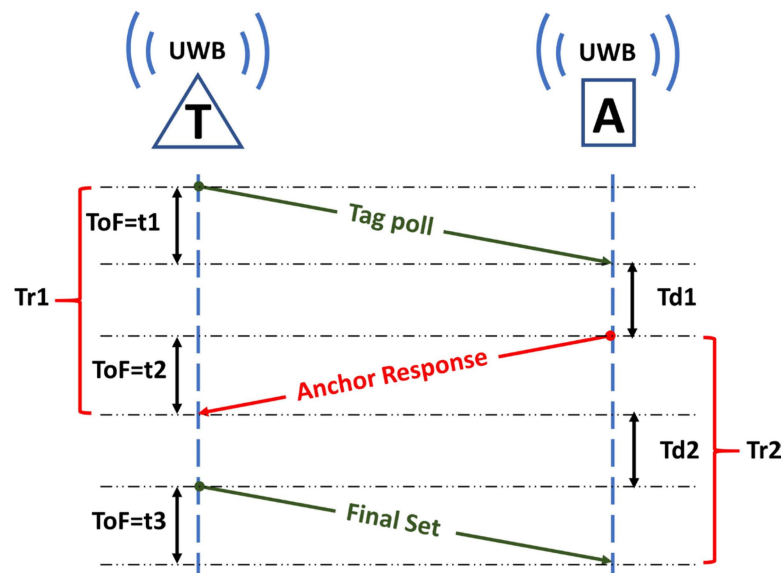


Figure 6. SDS-TWR tag–anchor signal propagation times.

3.2.2. Time Difference of Arrival (TDoA)

This algorithm determines the position by measuring the difference in the arrival time of the signal sent from the tag to the different anchors. The difference in distance from each anchor is calculated by the difference in arrival time at each anchor. This results in multiple hyperbolas passing through the tag's location. The intersection of these hyperbolas is the

tag's location. One of the advantages of this method is that there is no need to send a response message from the anchors to the tag, as the arrival time of the signal at each anchor is calculated separately, and then the difference between the arrival times is subtracted to determine the location of the moving tag. This saves processing time and increases the speed of the system. This method requires synchronization between anchors, which adds complexity to the implementation of this algorithm. However, this algorithm has great flexibility in the number of tags that can be tracked at the same time. Synchronization can be implemented using a direct wired connection between the anchors, which is more accurate. Synchronization can also be achieved wirelessly, and in this case, synchronization requires periodic calibration to obtain the required accuracy [15–18]. Figure 7 explains how the algorithm works. The tag begins sending a request to the neighboring anchors, depending on their location on the tag. After the signal reaches all three anchors, it takes different times to arrive at t_1 , t_2 , and t_3 . By calculating the difference between the three times, the location of the tag can be determined.

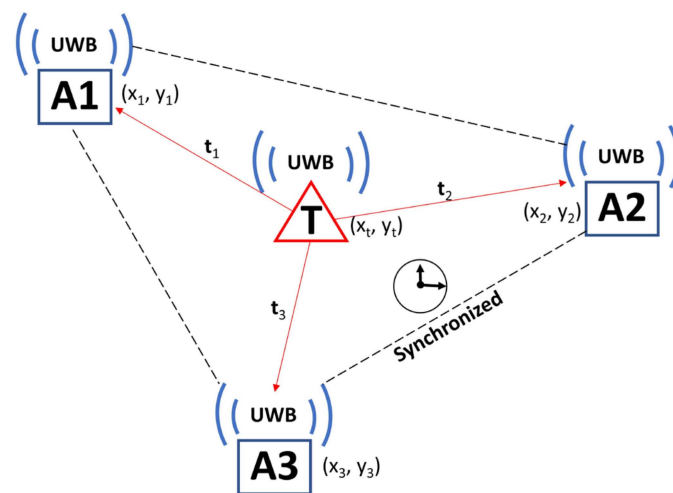


Figure 7. Topology of the TDoA positioning algorithm.

This method can be used differently, where the poll UWB signals are sent by the synchronized anchors, the different times of arrival for each anchor are calculated separately in the mobile tag, and the location is determined based on the difference between the different times of arrival for the anchors. This method is called reverse time difference of arrival (RTDoA).

3.2.3. Phase Difference of Arrival (PDoA)

This method is based on measuring the difference in phase between the ultra-wideband waves received from the transmitter (tag or anchor). Using this algorithm, the location can be found with the smallest possible number of anchors and tags by integrating two or more antennas (antenna array) to receive the signal in the receiver. This algorithm is used to estimate the angle of arrival or departure (AoA or AoD), as well as the distance between the sender and the receiver (ToA or ToD). In the previous methods, UWB units containing one antenna were used to calculate the time of flight and convert it into distance. In this method, a location can be found between a tag and an anchor, one of which contains two or more antennas with a distance d between them, without the need for additional system infrastructure. The recipient measures the phase difference between the signal received on each of the two antennas, and from this information, the algorithm can find the angle of the received UWB waves and determine the values of θ_1 and θ_2 as well as the distance between the sender and each antenna separately by calculating the time-of-flight values t_1 and t_2 , and this enables the location to be determined without the need for additional anchors. This algorithm is widely used for angle-of-arrival or angle-of-departure (AoA or AoD) estimation in IPSs. Figure 8 explains the structure of the PDoA algorithm [19–22].

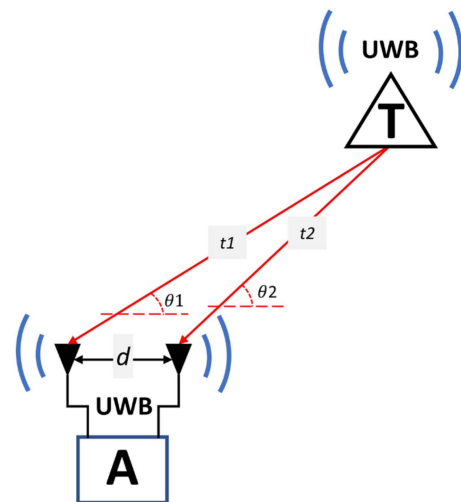


Figure 8. PDoA positioning algorithm.

3.3. Optimization Filters and Fusion-Based Algorithms for IPSs

Since the emergence of indoor positioning systems, and due to the margin of error in determining the correct location, important attempts have been made to reduce the error resulting from the delay time in the antennas or from the passage of waves through obstacles. The antenna delay time occurs when signals are exchanged between ultra-wideband units; a time delay occurs in transmission and reception as the signal propagating through its analog circuits suffers from delays in the transmitting and receiving antennas. Such delays can lead to range estimation errors between system units and, ultimately, affect the accuracy of real-time positioning systems unless these errors are measured and corrected [23]. Moreover, the delay resulting from the passage of signals through obstacles leads to an error in the accuracy of the system's performance. This causes incorrect measurements of the wave propagation time (ToF), through which the distances between the system units are calculated to determine the objects' location. Many methods, algorithms, and filters have been used to reduce this type of error. In the following sections, the most important of them are briefly explained.

3.3.1. Least Squares Algorithm (LS)

The least squares method is used to reduce the margin of error in indoor positioning systems by finding the closest path between the coordinates of the tag movement to be tracked. After obtaining the coordinates of the tag movement (x_t, y_t) from Equation (8) and the linear equation ($Ax = b$), the estimated coordinates can be calculated using the least squares method as follows:

$$A = \begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) \\ 2(x_3 - x_1) & 2(y_3 - y_1) \end{bmatrix}$$

$$x = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

$$b = \begin{bmatrix} d_1^2 - d_2^2 + x_2^2 + y_2^2 - x_1^2 - y_1^2 \\ d_1^2 - d_3^2 + x_3^2 + y_3^2 - x_1^2 - y_1^2 \end{bmatrix}$$

$$x = (A^T A)^{-1} A^T b \quad (11)$$

where A is an $m \times n$ matrix, x is an $n \times p$ matrix, and b is an $m \times p$ matrix. The " $[]^T$ " in Equation (11) represents the matrix transpose. Using Equation (11), the coordinates of the location of the guessed tag can be calculated depending on the provided coordinates of the anchors' location using the least squares method.

3.3.2. Kalman Filters (KFs)

The Kalman filter is a recursive estimation method that is typically used to calculate or predict the states of a system based on a model or noisy measurements of it considering additive white Gaussian noise. Its method of operation is based on predicting the next state based on information from the previous state. It calculates the values of the states of a dynamic system in an optimal way that minimizes the expected value of the square of the difference between the prediction and the correct state, and then uses them as inputs for the next iteration [24]. Many developments have been made to the regular Kalman filter to improve its operation, resulting in new types of this filter, the most prominent of which are the extended Kalman filter (EKF) and the unscented Kalman filter (UKF).

The limitations of the Kalman filter in dealing with primarily linear systems led to the development of the extended Kalman filter, which represents the nonlinear version of the Kalman filter. The unscented Kalman filter (UKF) was proposed by Goller and Ullmann in 1997. The central operation of the Kalman filter is to propagate a Gaussian variable (GV) through the dynamics of the system. In the EKF, the state distribution is approximated by the GV, which is then propagated through a first-order linear equation of the nonlinear system. This process can introduce large errors in the posterior mean and variance of the transformed GV, which can result in poor performance and deviations from the true value. The UKF is used to address this problem, using deterministic sampling. The state distribution is again approximated by the GV, which is now represented by a set of carefully selected sample points rather than the single average point in the EKF. The captured sample points contain the true mean and variance of the GV. When propagated through the true nonlinear system, the posterior mean and variance are accurately scaled to the third degree (Taylor series expansion) of the nonlinear state. This is not possible with the EKF, as it does not exceed first-class accuracy [25,26].

3.3.3. Particle Filter (PF)

Particle filter (PF) is another application of Bayesian filters that is used for position estimation or to solve state estimation problems. PF generates particles to represent the state of the system, where each particle represents a particular state of the position and direction of the tracked object (tag). When the object to be tracked starts moving, the filter generates particles with a normal distribution, where there is no prior information about the location of the tag. The coordinates and orientations of the tag are generated and distributed according to the geometry of the location. This process is called sampling, where each particle has its own weight parameter (number). This number represents the probability of the system being in a particular location or state, and the larger the number (weight parameter), the greater the probability of finding the tag in the relevant location. Then, as the tag continues to move, the measurement data captured by the sensors provide more accurate information about the location of the tag being tracked [27,28].

4. Commercial UWB Modules and Chips for IPSs

Many companies that produce and market real-time location and indoor positioning systems use off-the-shelf ultra-wideband modules to reduce the time, cost, and complexity of manufacturing these modules themselves. In this section, we will discuss the most popular commercial ready-made units that are used in indoor positioning systems.

4.1. Qorvo DW1000 UWB Family

The Qorvo (Previously DecaWave) DW1000 (Qorvo Inc., Greensboro, NC, USA) is one of the first ultra-wideband units that contains all basic circuits for the transmission and reception process in one chip. It began to spread in the middle of the last decade and is still widely used today, despite the emergence of modern competing products. This product provides different attractive features that enable its widespread use, such as low cost, measurement accuracy of up to 10 cm in line-of-sight conditions, low power consumption compared to cellular radio modules, working on six channels within the radio frequencies

3.5 to 6.5 GHz, data transfer capacity of up to 6.8 Mbps, and the ability to communicate over distances of more than 200 m in the case of line-of-sight (LoS) communication. It can be used for both two-way ranging and TDoA algorithms for IPSs and can be hosted by any processor with an SPI interface [29–31].

4.2. Qorvo DW3000 UWB Family

The DecaWave DW3000 transceiver module (Qorvo Inc., Greensboro, NC, USA) is the advanced version of the previous family DW1000 [32]. The module is an IEEE 802.15.4-2011 [33] and IEEE 802.15.4z [34] UWB module based on the Qorvo DW3000 chips, which include four different versions (3110, 3120, 3210, and 3220) depending on the package and the AoA/PDoA measurement capability. It is a fully integrated UWB transceiver with a ceramic antenna, clock circuit, power management, and all required RF circuits in one module. The module has high multipath fading immunity and can be used in two-way ranging (TWR) or time difference of arrival (TDoA) location algorithms to execute real-time tracking and positioning tasks. It supports channels 5 and 9 (6489.6 MHz and 7987.2 MHz) and data rates of 850 kbps and 6.8 Mbps. This module can be integrated with any host processor for IPS implementation and supports SPI interfaces of up to 38 MHz. It is interoperable with the Apple U1 and U2 chips and supports high tag densities in real-time location systems.

4.3. NXP Trimention UWB Chips Family

The Trimention UWB family (NXP Semiconductors, Eindhoven, The Netherlands) has many products that are well customized to suit different applications. Trimention SR150 is used for UWB tags and anchors in IoT applications, and it is ideal for applications with large-scale infrastructure. It has a pre-installed FiRa™ stack (which stands for “fine ranging”), which has the integrated FiRa™ MAC, supporting interoperability with a set of UWB devices in the market. It adds angle-of-arrival (AoA) technology for high-precision applications, and it is interoperable with IEEE 802.15.4z higher-pulse-repetition UWB. Other members of the Trimention family of UWB chips, such as SR150, SR040, OL23D0, NCJ29D5, and NCJ29D6, can be used with different technologies and applications [35,36].

4.4. Microchip ATA8350/ATA8352 UWB Transceivers

The Microchip ATA8350 UWB transceiver (Microchip Technology Inc., Chandler, AZ, USA) is designed for measuring hardware-controlled sequential distance for localization. It is compliant with UWB regulations and supports a frequency range of 6.2–7.8 GHz with a data rate of 246 Kbps. It can be hosted by any processor with an SPI interface of up to 20 MHz and can measure the ToF with a resolution of ± 15 cm. The ATA8352 transceiver has a wider frequency range of 6.2–8.3 GHz, transfers higher data rates of up to 1 Mbps, has SPI control with up to 24 MHz, and supports the time difference of arrival (TDoA) approach. It has a more accurate ToF measurement resolution of ± 4.5 cm, which reflects better system accuracy [37].

4.5. Infineon 3DB6830 UWB IC

The Infineon 3DB6830 (Infineon Technologies Switzerland AG, Zurich, Switzerland) is a low-power IEEE 802.15.4f [38]-compatible UWB transceiver with an integrated security layer that makes it more suitable for distance measurement for security applications. The IC measures the distance based on direct ToF measurement and can be driven by any host processor or microcontroller that can execute random number generation and authentication routines. It works within the upper UWB bands from 6 to 8 GHz. It can cover up to 120 m in LoS communication conditions, with ± 10 cm accuracy [39].

4.6. SPARK SR1010/SR1020 UWB Transceiver

The Spark SR1010/SR1020 (Spark Microsystems International Inc., Montreal, QC, Canada) is a low-power UWB transceiver suitable for energy-efficient communications with ultra-short latency and supports a reconfigurable frequency range of 3.1–6 GHz for

SR1010 and 6–9.3 GHz for SR1020. The transceiver’s data rate can be contentiously scaled from 1 Kbps to 10 Mbps to achieve ultra-low power consumption over a wide range of data rates, which makes it a flexible solution for many applications. These chips can measure the distance by measuring the time of flight of the generated UWB signal between two chips, with a range approaching 100 m in LoS conditions, and with a measurement accuracy of about ± 30 cm [40].

4.7. B-UWB-MOD1

The B-UWB-MOD1 (STMicroelectronics, Geneva, Switzerland) is an ultra-wideband module designed for implementing high-precision indoor positioning and location systems. It is an ultra-compact surface-mounted module that is designed to work with UWB frequencies from 3.25 to 4.75 GHz and support the UWB channels 1, 2, 3, and 4 with bandwidths of 500 MHz and 1 GHz. It can cover ranging distances of up to 600 m in LoS conditions with 10 cm distance measurement accuracy and can provide an adjustable sample rate of up to 250 per second. The data can be acquired using the UART communication bus. This module is currently used as the core element of the commercial evaluation kit B-UWB-MEK1 (STMicroelectronics, Geneva, Switzerland), which is used for the prototyping, evaluation, and demonstration of indoor location systems [41].

4.8. Summary of the Commercial UWB Modules and Chips for IPSs

In this section, we present a summary of the most prominent characteristics of some available commercial chips and units available on the market that can be used in the design and development of real-time indoor positioning systems operating with ultra-wideband technology, as shown in Table 1.

Table 1. Commercial UWB modules and chips for IPSs.

Vendor	Chip/Module	Estimation Method	LoS Accuracy ($\pm$ cm/ $^\circ$)	Operation Range (m)	Frequency Range (GHz)	UWB Channels	Application	Reference
Qorvo	DWM1000	TWR, TDoA	15 in 2D, 30 in 3D	300	3.5–6.5	1–5, 7	IPS, asset, employer	[42]
Qorvo	DW3110, DW3210	TWR, TDoA	15 in 2D, 30 in 3D	50	6.5, 8	5, 9	IPS, asset, employer	[32,43]
Qorvo	DW3120, DW3220	TWR, TDoA, PDoA	15 in 2D, 30 in 3D	50	6.5, 8	5, 9	IPS, asset, employer	[44,45]
NXP	SR150	AoA, TWR, TDoA	10/3	50	6.24–8.24	5, 6, 8, 9	Industrial IoT, RTLS	[35]
NXP	SR40	TWR, TDoA	10	50	6.24–8.24	5, 6, 8, 9	IoT battery-based, IPS	[46]
Microchip	ATA8350	TDoA	15	-	6.2–7.8	5–8	IoT battery-based	[47]
Microchip	ATA8352	TDoA	4	-	6.2–8.3	5–9	IoT battery-based	[48]
Infineon	3DB6830	TWR	10	120	6–8	5–9	Industrial IoT	[39]
Spark	SR1010	TWR	30	100	3.1–6	1–5	IoT applications, Asset and individual tracking	[49]
Spark	SR1020	TWR	30	100	6–9.3	6–9	Industrial IoT	[49]
STM	MOD1	TWR	10	600	3.25–4.75	1, 2, 3, 4	RTLS, asset, employer	[41]

4.9. Commercial UWB-Based IPSs

The growing need and demand for indoor positioning systems has led many companies, vendors, and factories to focus on manufacturing and developing commercial systems that suit market requirements. A complete package is developed that includes the hardware components and compatible software enabling the system to display the tasks required to be performed optimally. Table 2 reviews selected providers of indoor positioning systems operating with ultra-wideband technology.

Table 2. Commercial UWB-based solutions for IPSs.

Vendor	Algorithm/Topology	Accuracy ($\pm$ cm)	Operation Range (m)	Frequency Range (GHz)	Application	Reference
Inpixon	TWR/AoA	40	50	6.4	Asset, employer	[50]
Eliko	TWR/TDoA	50	70	NA	RTLS	[51]
Litum	TWR/TDoA	NA	70	3.25–6.75	Asset, employer, forklift	[52]
WOXU	ToF/TDoA	10	NA	3.24–6.74	RTLS	[53]
Sewio	TDoA	30	50	3–7	Asset, employer, forklift	[54]
KKM	TDoA	30	NA	NA	Asset	[55]
Kinexon	TDoA/AoA	20	NA	6.25–6.75	RTLS	[56]
Pozyx	ToF	30	50	3.5–6.5	People, assets, vehicles	[57]
Zebra	ToF	30	200	6.35–6.75	Transportation, warehouse	[58]
Tsingoal	ToF/TDoA	30	NA	NA	RTLS	[59]
Ubisense	TDoA/AoA	30	65	6–7	Asset	[60]
Kathrein	ToF/TDoA	10	40	3–10	Vehicle, forklift	[61]
Ubudu	TDoA/TWR	20	NA	NA	Portable tracking device	[62]
Tracktio	TDoA	30	150	3–7	Worker, warehouse	[63]
Leisure	ToF/TDoA	1	100	3.1–10.6	Vehicles	[64]
MK	TDoA/AoA	7	NA	6.5–8	RTLS	[65]
BeSpoon	TWR	10	880	3.99	RTLS	[66]

5. Real-Time UWB Indoor Positioning Systems (RTIPSs)

The urgent need to develop real-time indoor positioning systems has emerged with the great developments in the sciences of automation engineering and robotics in particular, and in the sciences of electrical engineering and informatics in general, to provide the possibility of accurate tracking of self-driving systems such as robots and drones in closed environments, i.e., inside buildings. The development of real-time tracking and indoor positioning systems faces many challenges that affect the performance of the systems and limit the required accuracy.

5.1. Obstacles and Non-Line-of-Sight Obstructions

Calculating the wave propagation time with high accuracy leads to determining the location with high accuracy, but the time calculation may be subject to errors that cause major defects in the system's performance in general; this is especially the case inside buildings, where many sources of interference and obstacles cause defects in the calculation of the wave propagation time. An error in calculating the real distance leads to an error in determining the real location of the object to be found or tracked.

Fading is one of the common phenomena that affect the performance of ultra-wideband communication systems. It occurs when the signal propagates in multiple paths, which causes signal fluctuation that affects the strength and quality of the UWB signal, leading to errors in time-of-flight calculation and increasing the system error rate.

Shadowing is another common and more popular phenomenon that affects the performance of IPSs. It occurs when the UWB signals are attenuated or blocked by obstacles in the indoor environment [67]. The fading and shadowing effects are critical parameters to be considered during the system performance and error analysis as part of the UWB-based indoor positioning wireless communication system [68]. One of the most important problems affecting the accuracy of measuring propagation time is the presence of obstacles in the path of direct wave propagation between two points, which is called the non-line-of-sight (NLoS) problem. In this case, there are two possibilities: The first is for the wave to penetrate the obstacle and complete its path to the target point. In this case, the error in measuring the distance occurs due to the delay resulting from adding the penetration time. This increases the time taken to reach the target, which leads to an increase in the calculated distance and, thus, an error in determining the location. In the second possibility, the direct wave is unable to penetrate the obstacle; thus, it is reflected or deflected. In this case, the reflected or deflected indirect waves may reach the target, traveling a greater distance by bouncing

off walls or obstacles, which adds a delay time that may be greater than in the first case, resulting in a greater error in determining the location of the target to be tracked [69,70]. Figure 9 explains the NLoS problem in UWB-based IPSs.

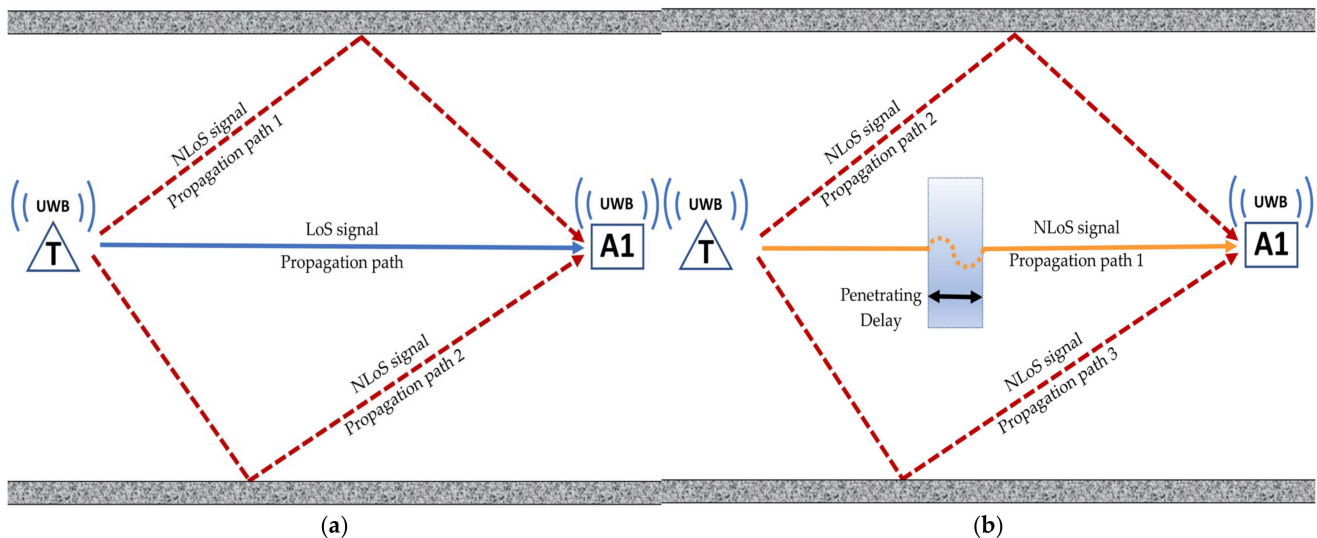


Figure 9. Signal propagation path for UWB IPSs in (a) LoS and (b) NLoS.

Many researchers have suggested different solutions and methods to address errors caused by the non-line-of-sight problem. Yang et al. [71] proposed an efficient cooperative positioning algorithm to mitigate the environmental noise and enhance the performance and accuracy of IPSs in non-line-of-sight environments. The algorithm is based on selective tags' coordinate information sharing, and it includes three steps: initial positioning, tags classification, and precise positioning. The evaluation process is conducted in a non-line-of-sight environment, and the evaluation results show a 71.7% accuracy improvement. He et al. [72] proposed an approach that combines Discrete Wavelet Transform (DWT) and convolutional neural networks (CNNs) to solve non-line-of-sight ultra-wideband signal recognition. The UWB signal coefficients are extracted from the Channel Impulse Response (CIR) using DWT, and then the coefficients are fed as inputs to a CNN to explore deep features to distinguish between NLoS and LoS signals. The experimental results revealed that this method achieves better accuracy compared to other approaches. Qin et al. [73] proposed a technique for compensating the NLoS occlusion errors of the UWB IPS signals using a micro-electromechanical system (MEMS) inertial measurement unit (IMU)-assisted UWB localization. The NLoS occlusions of UWB devices are identified with the channel information of the UWB transceivers and tag–anchor distance variation patterns. The localization process is implemented by fusing the UWB and IMU data with an unscented Kalman filter (UKF) framework. The data fusion of the system is adjusted according to the NLoS effect on the UWB signals. System tests have shown high efficiency in identifying and compensating for errors caused by NLoS in the used UWB IPS modules. Su et al. [74] proposed a system (UWBLoc) to mitigate the NLoS errors in IPSs using ranging residuals to characterize the influence of the environment on ranging and measuring distance and angles using the Taylor-series-based least squares (TS-LS) method. Similar approaches have been proposed in [75–78].

5.2. Real-Time Implementation Ranging and Topologies

The unique properties of ultra-wideband waves have supported their widespread use in real-time positioning systems. As explained in the previous paragraphs, several companies have introduced products that support the direct development of tracking and positioning systems using different transceivers and modules based on ultra-wideband technology. Below are the most prominent research works focusing on indoor tracking and

positioning systems using ultra-wideband technology and completed during the survey period described above, which represents the main contribution of this work.

5.2.1. Two-Way Ranging-Based Indoor Positioning Systems

This method is one of the common methods for implementing real-time tracking and positioning systems. It can be implemented using two topologies: The first uses a reference anchor to calculate the distances between the ultra-wideband units and send them to the monitoring and display server via a cable or wireless connection. The second approach calculates the distances between the moving tag and the active anchors in the moving tag itself and sends the data via the cloud to the monitoring and display server. Figure 10 shows these two topologies [8].

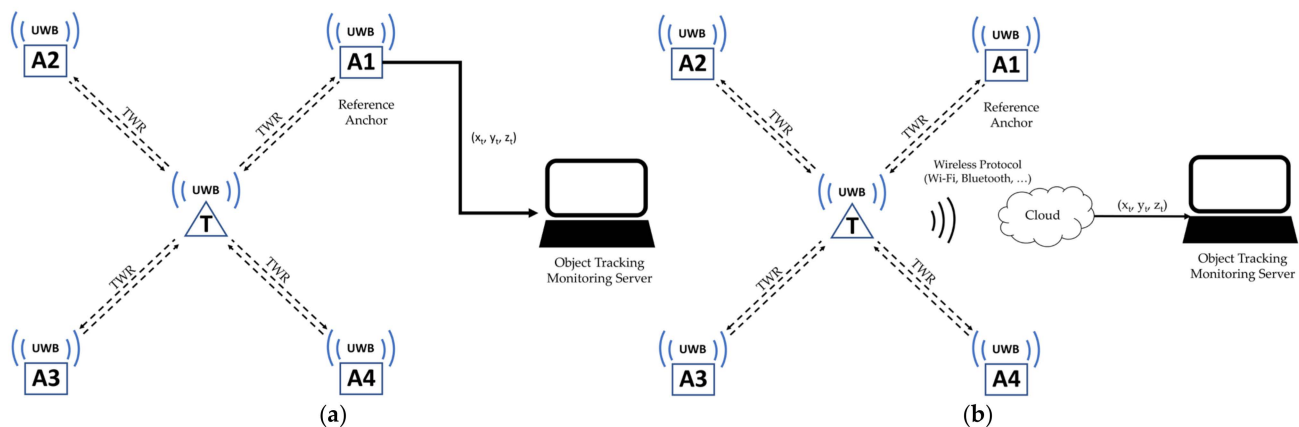


Figure 10. TWR IPS topologies: (a) anchor-based; (b) tag-based.

Many tracking and positioning systems have been implemented using this method. He et al. [79] proposed a single-sided two-way ranging (SS-TWR) positioning system based on filtering and clock compensation. The system uses the Qorvo DW1000 (Qorvo Inc., Greensboro, NC, USA) UWB modules for both tags and anchors. The proposed system uses the anchor-based topology and consisted of three anchors and one tag. The reference anchor is connected by wire to the server laptop using a serial port. The random forest algorithm estimates new distances based on previous SS-TWR algorithm values. The results of both random forest and SS-TWR algorithms have been evaluated using the root-mean-square error (RMSE), and the results showed better performance for the random forest algorithm, with an improvement of approximately 4.85 cm compared to SS-TWR. Al-Okby et al. [80] proposed a low-cost multi-tag IPS for object tracking in indoor environments. The system's UWB hardware uses the MaUWB_DW3000 (Makerfabs Corporation, Shenzhen, China) module, which includes the new Qorvo DW3000 UWB transceiver. The system consists of three fixed anchors and can track up to 16 movable tags. The trilateration technique with the least squares method has been used for system implementation and testing in both LoS and NLoS situations. All UWB measurements from the system's UWB units are accumulated by the reference anchor and forwarded to the control and monitoring server laptop via a COM port. A Python-3.7 program is used for monitoring the movable tag based on the map of the testing area. The practical tests indicate that the system can achieve an accuracy of 50 cm. Cheng et al. [81] proposed the design of an IPS based on TWR and trilateration algorithms. The system consists of four anchors and one movable tag. All of the units have similar hardware designs, including the Qorvo DW1000 UWB modules for distance measurements and the STM32F103ZET6 (STMicroelectronics, Geneva, Switzerland) microcontroller for data preprocessing and UWB control tasks via an SPI communication bus and powered by a lithium battery. The experimental tests proved a low error rate of approximately 14 cm, and the authors suggested this system for underground coal mine personnel. Le Minh et al. [82] proposed a double-sided two-way ranging (DS-TWR)-based positioning system with a Kalman filter for improving the system accuracy.

The system used the Qorvo DW1000 UWB modules for system implementation, with four fixed anchors and one movable tag responsible for the distance calculation with all ultra-wideband units and communication with the monitoring station (tag-based topology). The use of the Kalman filter enhances the system's performance and decreases the distance measurement errors in both LoS and NLoS environments. Qi et al. [83] used a derivative unscented Kalman filter (DUKF) and root-mean-square error (RMSE) to estimate and reduce the nonlinear error of anchor positions in a TWR-based IPS. The system used the Qorvo DW1000 UWB modules and micro-electromechanical system (MEMS) inertial sensor for real-time implementation with the anchor positions' error calibration, which improved the localization precision and robustness of the system. Sidorenko et al. [84] investigated the multiple error corrections of clock drift between the reference anchor and the tag for the DecaWave DW1000 transceiver (Qorvo Inc., Greensboro, NC, USA) and analyzed the most common TWR protocols in real-time implementations. The test results showed that alternative double-sided two-way ranging (AltDS-TWR) has the best results, and that it can correct the signal power and hardware delays for every UWB unit individually. The used UWB unit was the Qorvo (DecaWave) EVK1000, which consists of the DW1000 IC with the ST32F105 (STMicroelectronics, Geneva, Switzerland) microcontroller and has a built-in LCD, various interfaces (JTAG, SPI, USB), and a wideband planar omnidirectional antenna. Laadung et al. [85] proposed two general active-passive two-way ranging (TWR) methods for UWB-based IPSs: The first (AP1-TWR) uses the UWB packet exchange and listens to the other anchors' transmissions, with knowledge about the anchors' locations. The second (AP2-TWR) only listens to the other anchors' transmissions. The experimental tests were conducted using the Eliko UWB RTLS (OÜ Eliko Technology, Tallinn, Estonia), which is based on the DecaWave DW1000 module. Five fixed anchors and one tag were used, and the distances between the anchors and the anchor coordinates were measured using the DISTO S910 (Leica Geosystems AG, Heerbrugg, Switzerland) laser distance meter. Both methods were tested with the SS-TWR, SDS-TWR, and AltDS-TWR active ranging methods. The results indicated that the AP2-TWR method consistently outperforms the AP1-TWR method by about 10 to 20%. Herbruggen et al. [86] proposed a real-time approach for optimal anchor node selection in TWR. Instead of ranging with all nearby anchors, including those with poor channel characteristics, the new method allowed for the selection of the best-performing anchors based on the best link quality. This new approach enhances the accuracy by up to 15 cm, which represents a 50% improvement compared with traditional TWR. The practical tests were implemented using the Wi-Pos UWB open-source platform, which includes the Qorvo DW1000 transceiver with wireless backbone. Other TWR-based IPSs have been proposed in [87–97].

5.2.2. Time Difference of Arrival (TDoA)-Based Indoor Positioning Systems

This is a flexible and scalable method for determining the location of moving tags by recording the difference in the arrival time of the signal between active anchors that must be synchronized. The advantage of this method is its low power consumption, because the identification wave needs to be sent only once from the tag to the anchors, without the need for a response or confirmation wave. The timestamps recorded at all anchors are collected by the reference anchor or a gateway and forwarded to the tracking monitoring server, where the location coordinates are calculated and the desired movement of the tags is tracked.

It is also possible to implement this method in reverse, called reverse time difference of arrival (RTDoA). The anchors send synchronized blinks at specific time intervals in the active tracking area. When the moving tag enters the specified area, it receives the blinks and calculates the distances between it and the anchors using TDoA. All of the calculated distances are then forwarded to the monitoring server for implementing real-time tag monitoring using multilateration. Figure 11 shows the two TDoA topologies [8].

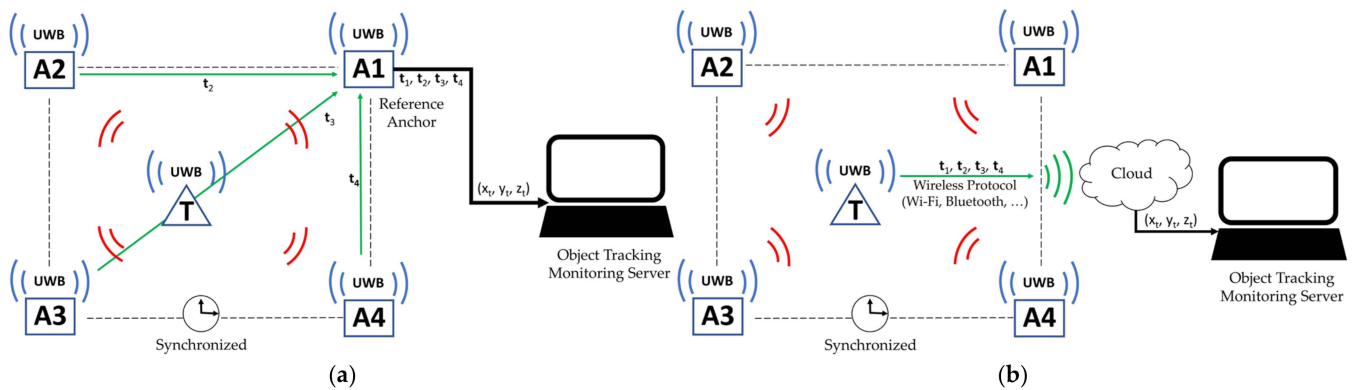


Figure 11. TDoA IPS topologies: (a) TDoA; (b) RTDoA.

This method has been widely used in the implementation and development of indoor tracking and positioning systems due to its unique advantage of saving energy consumption, especially since the targeted systems use batteries extensively in their operation. Many studies, system developments, system tests, and evaluations for TDoA have been proposed. Shyam et al. [98] proposed a TDoA-based RTTS for patient tracking in healthcare environments. The system consists of four anchors and two wearable tags embedded in a wristband. The anchors are synchronized, fixed in a wall, and connected through Ethernet via a 24-port Ethernet hub. The nearest anchor for the tags is considered to be the reference anchor, and it is responsible for collecting the TDoA information. All of the data are forwarded to the system server, which is a Raspberry Pi platform (Raspberry Pi Ltd., Cambridge, UK). The server is responsible for calculating the coordinates of the movable tags. The Qorvo DW1000 UWB module has been used in both anchors and tag design, where the STM32F407VET6 and the STM32L041K6U6D (STMicroelectronics, Geneva, Switzerland) microcontroller have been used for anchors and tags, respectively. A mobile-phone- and web-based application can be used for monitoring. Liu et al. [99] proposed a high-precision wireless IPS based on TDoA combined with the Taylor algorithm. The system's UWB units have the same hardware design and can be programmed as anchors or tags. The unit's hardware consists of the Qorvo DW1000 and the STM32F103RCT6 (STMicroelectronics, Geneva, Switzerland) microcontroller, which communicates via an SPI bus. The system testing was performed using four fixed synchronized anchors and one movable tag. The test results revealed that the system's positioning measurement errors do not exceed 10 cm.

The main challenge in implementing TDoA-based IPSs is the need for time synchronization of the used anchors. One possible solution to overcome this challenge is to use a synchronization reference anchor that broadcasts periodic synchronization signals to the other anchors. Several researchers have developed the TDoA method in a way that does not require time synchronization between anchors, reducing the complexity and cost. Xue et al. [100] proposed a deep learning algorithm for TDoA IPS security with missing data and measurement errors called "DeepTAL". First, the algorithm is trained with the system without the anchors' synchronization condition, and then it is applied to the TDoA synchronized system in real time. The system was implemented using the Qorvo DW1000 UWB radio transceiver with an ARM cortex M43 STM32f105 (STMicroelectronics, Geneva, Switzerland) microcontroller. The practical tests of the system were implemented using four anchors and one tag. The test results show that the system can predict the TDoA accurately in the presence of measurement errors or missing data. Cho et al. [101,102] proposed a 3D TDoA-based IPS for tracking the workers in hazardous indoor areas of construction sites. The system hardware consists mainly of the commercial Samsung SmartTag+ (Samsung Electronics Co., Ltd., Suwon, Republic of Korea) module. The UWB chip in this tag is the NXP Trimension SR040 (NXP Semiconductors, Eindhoven, The Netherlands), which is powered by a small coin-cell battery and used for tracking objects using new Samsung

smartphones with UWB technology. Four SmartTag+ modules were used as fixed anchors for indoor environments, and a Samsung smartphone was used as a mobile tag. The system's testing showed that the spatial configuration of the anchors and tags with increasing height differences has a significant impact on the accuracy of the 3D indoor localization. The system tests demonstrated an accuracy of 35 cm. Patru et al. [103] proposed a scalable TDoA-based IPS "FlexTDOA". The system topology is based on passive tags and active fixed anchors, where the anchor clocks' synchronization is not required; instead, the flexible time-division multiple access (TDMA) scheme is used for the anchor communications. The system hardware was built using the Qorvo DW3000 UWB transceiver with the ARM cortex M4 STM32F429ZIT6 (STMicroelectronics, Geneva, Switzerland) microcontroller for control and data processing. The ranging data were transmitted using a USB 2.0 port, and each unit had the main communication buses, such as UART, SPI, and I2C. The system units used a rechargeable Li-ion battery as a power source and the MCP73830 IC for power management. The UWB unit was configured to operate on channel 5, with 6.5 GHz and a data rate of 6.8 Mb/s. The system was tested using ten anchors and one tag. The test results show that the proposed method has better performance than classical TDoA, with an accuracy of ≈ 17 cm in LoS and ≈ 22 cm in NLoS environments. Bottigliero et al. [104] proposed a low-cost TDoA RTLS that does not require time synchronization to reduce the system complexity by using a one-way communication scheme. The tags blink a sequence of 2-nanosecond pulses on a 7.25 GHz carrier frequency, which is received by the anchors with a two-step correlation analysis to calculate the ToA of the tag at each anchor. The hardware of the system has a custom design for the anchors and the tag. The system tests used three anchors and one tag, and the achieved accuracy was ≈ 10 cm. Pérez-Solano et al. [105] proposed another approach for unsynchronized TDoA IPSs using one-way ranging for time-of-arrival calculations. The system consists of hardware and software parts; the system units include the Qorvo (DecaWave) DWM1000 UWB modules with a 2-millisecond packet transmission time and the Raspberry Pi platform (Raspberry Pi Ltd., Cambridge, UK) for control and processing. The system was tested in two different deployments: the first in a single room, with an accuracy of ≈ 15 cm, and the second in a multi-room environment, with 56 test points and an accuracy of ≈ 50 cm. Similar TDoA-based IPSs have been proposed in [106–109].

5.2.3. Phase Difference of Arrival (PDoA)-Based Indoor Positioning Systems

This method is significant for peer-tracking applications, especially for mobile tracking applications that have to be implemented without relying on a fixed infrastructure. Positioning is determined by calculating the distance and direction of the UWB nodes. Measuring the angle of arrival (AoA) and the time of flight between tags and anchors can be achieved using both angle-of-arrival and two-way ranging algorithms [8]. One or both of the used UWB units should have two antennas to allow for the AoA calculations. Many researchers have used PDoA for IPS implementation. Zhang et al. [110,111] proposed a PDoA-based IPS for tracking the elderly in smart homes. Their work aimed to overcome the NLoS problem in elderly living environments. The coarse location was calculated initially to find the nearest LoS anchors in the NLoS environment, and then the NLoS anchors' measurement was canceled and only the LoS anchors were used for the elderly location tracking. The system hardware used two different units from Qorvo (DecaWave Limited, Dublin, Ireland): a UWB transceiver, and a PDoA chip combined with a host processing board. The tag had a different design, and it was included in a wearable wristwatch with GPS, Zigbee, and some inertial and biosensors. The system was practically tested with nine fixed anchors in different positions, with LoS and NLoS positions for the tag, and the results showed that the system performed the tracking tasks successfully. Botler et al. [112] proposed four different machine learning models for PDoA IPSs. The models were trained with real-time data from commercial UWB units. The system hardware consisted of the DecaWave Beta-PDoA UWB kit, and the practical testing of the developed models showed that the random forest regressor model had the best performance, with a mean param-

ter of 8° . Ge et al. [113] proposed a single-anchor RTLS using wrapped PDoA based on an antenna array with arbitrary geometry. The object localization was implemented by measuring the ToF and the wrapped phase of the UWB signals. The localization process included obtaining parameters by running a soft positional information algorithm, which was approximated by the Gaussian mixture model (GMM) to accelerate the filtering process. The system was practically implemented in a single-antenna tag and an eight-antenna anchor connected to the DecaWave DW1000 UWB transceiver, which could calculate the time of arrival and the carrier phase—the main parameters required for implementing PDoA. The system testing results revealed that localization accuracy of ≈ 8 cm could be achieved. Other PDoA approaches are presented in [114–120].

5.3. Indoor Positioning Systems for Robotics Applications

The indoor positioning systems of mobile robots are essential and indispensable, especially in swarm robotics, simultaneous localization and mapping (SLAM), and large-scale applications. From the perspective of functionality and practicality, the UWB-based IPS is more suitable than other technologies for robotics applications [121]. In this section of the paper, we will review the most important integrated works and research in this field. We consider the systems that contain all of the main parts of RTTs, such as the hardware components that allow the indoor tracking and positioning operation with the UWB technology, along with the interactive software to show and monitor the relative location information and object coordinates in a graphical user interface. The integrated system can simulate the physical reality in real time in the software, enabling real-time tracking with the best possible accuracy. Figure 12 gives an example representing the real-time tracking system in a Python-based monitoring program.

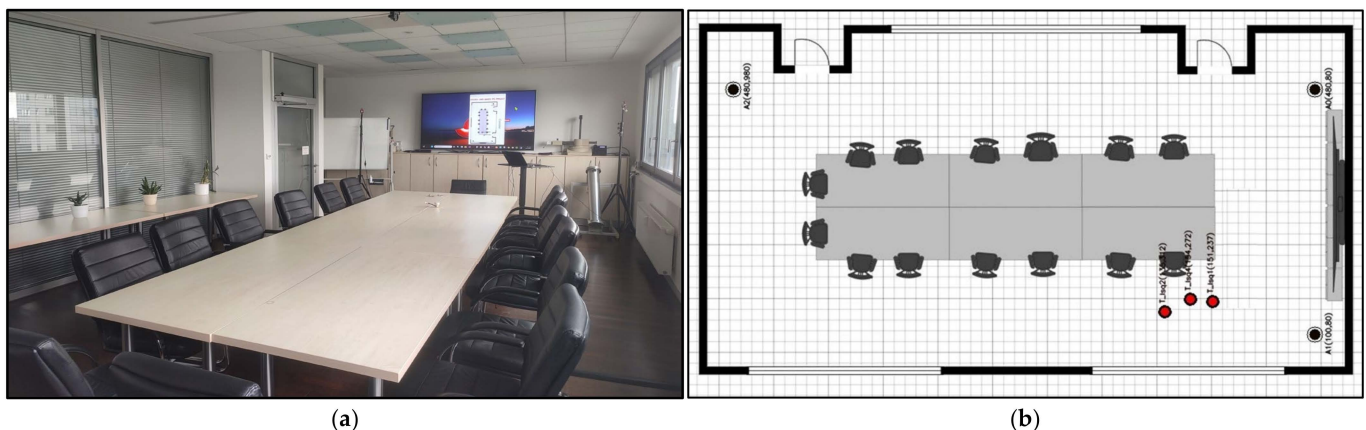


Figure 12. IPS: (a) real-time testing environment; (b) 2D simulated environment with real-time object tracking using Python [80].

There are many works related to this topic. Xu et al. [122] proposed an extensible RTLS for mobile robots. The system consisted of one tag fixed on the mobile robot, five anchors (four fixed and one with an unknown position), a camera, and an IMU. The UWB anchors did not require clock synchronization. This system can be extended by adding a new anchor, where the recursive least squares (RLS) approach is used for the new anchor's location estimation, and the maximum correntropy Kalman filter (MCKF) is used to fuse the data from the UWB network and IMU. The system's main hardware included the DecaWave DWM1000 transceivers for the UWB tag and anchors with a small mobile robot equipped with the KY-INS110 IMU unit (Beijing BDStar Navigation Co., Beijing, China), and MYNT EYE D1000-IR-120 stereo camera (MYNT AI, Jiangsu, China). The system tests indicated a 0.137 m root-mean-square error (RMSE) in the positioning location. Li et al. [4] proposed a UWB-based RTLS for mobile mining robots in harsh coal mine environments. The system used both UWB technology for location and IMU sensors for orientation to enhance its positioning accuracy. Four anchors and one tag were used to perform a 3D

positioning in the coal mine using the P440 UWB module (TDSR LLC., Petersburg, TN, USA). The MTi-G-710 motion tracker IMU module (Movella Inc., Henderson, NV, USA) was used to provide the robot's orientation data. The UWB tag and the IMU modules were fixed on the mobile tank robot. The data of both the UWB and IMU modules were fused using the error-state Kalman filter (ESKF). The system tests indicated that a localization accuracy of ≈ 20 cm can be achieved. Cano et al. [123] proposed a Kalman filter-based IPS for mobile robots. Their work aimed to mitigate the multipath issue of the UWB waves using the M-estimation robust Kalman filter (M-RKF) and leveraging an adaptive empirical variance model. The system hardware used the Qorvo DW1000 modules for both the anchors and tag, where the tag was fixed in a mobile ground robot surrounded by four fixed anchors. The TWR protocol was used for range measurements. The movable tag acquired all of the range measurements at a 280 Hz refresh rate. The practical system tests indicated that the use of M-RKF enhances the system localization performance by 14% in LoS and 31% in NLoS environments. Ranjan et al. [124] developed an IPS for mobile robots by integrating the filtering methods moving average (MVG), Kalman filter (KF), and extended Kalman filter (EKF) with a low-pass filter (LPF). The used hardware included the commercial POZYX UWB positioning system (POZYX, Ledeberg, Belgium) and the open-source ROS-based TurtleBot3 mobile robot (ROBOTIS Inc., Tokyo, Japan). The system testing showed that the extended Kalman filter with a low-pass filter (EKF + LPF) had the lowest root-mean-square error (RMSE) of 40.22 mm on the x-axis, 78.71 mm on the y-axis, and reduced mean absolute error (MAE) percentages of 3.39% for square paths, 4.21% for circular paths, and 6.16% for free paths. Tang et al. [125] proposed an enhanced IPS using human–robot collaboration. This method aims to reduce the number of fixed anchors in classical IPSs by using the mobile robot as a reference anchor of positioning systems, while also enhancing the positioning accuracy using inertial navigation data from pedestrian smartphones. The system was tested in three scenarios, the first with only three fixed anchors and pedestrians holding smartphones. In the second scenario, a mobile anchor (mobile robot with UWB module) was added, which added the LiDAR positioning information to enhance the system's performance. In the third scenario, only two mobile anchors and two fixed anchors were used. The test results indicated that the best system performance was achieved in the second scenario, with an RMSE of 0.355 m. The authors did not provide technical details about the used hardware components. Alghuraid et al. [126] developed a UWB-based IPS for grid-confined robotic platform position control. The system hardware used the commercial Qorvo MDEK1001 kit, which mainly uses the Qorvo DW1000 UWB transceiver. Eight modules were used as fixed anchors, one was used as a movable tag attached to a Mecanum four-wheel drive robot, and one was used as a listener for providing the streaming coordinates of the UWB system to the monitoring laptop via a USB COM port. The mobile robot was controlled by the Atmel ATmega2560 (Atmel Corporation, San Jose, CA, USA) microcontroller, and the robot communicated with the control station wirelessly using the HC-12 Bluetooth module (Guangzhou Huicheng Information Technology Co., Ltd., Guangzhou, China). The open-source Tera Term terminal software, Arduino IDE (Arduino LLC., Boston, MA, USA), and MATLAB (MathWorks Inc., Natick, MA, USA) were used for data processing and monitoring. The practical tests of the system achieved an X-Y location accuracy of 7 cm. Other robotics-based IPSs are presented in [127–135].

5.4. Indoor Positioning Systems for Unmanned Aerial Vehicle Applications

The use of tracking systems for unmanned aerial vehicles is an essential element. This is due to the nature of the tasks that they perform through flight and their specific applications, which are often in open outdoor environments outside buildings and usually use satellite navigation and tracking systems. The need to use these vehicles in indoor environments for tasks that mobile robots cannot perform is a challenge for them, as satellite signals are weak and unusable. Here, the need to use indoor tracking and positioning systems of various types, especially those operating on ultra-wideband waves, arises. The

current review paper is limited to indoor tracking and positioning systems, so this chapter will be limited to works that meet this description.

Kefferpütz et al. [136] presented an indoor navigation system for UAVs for fusing data from UWB, IMUs, and barometers using an error-state unscented Kalman filter. The Crazyflie 2.1 (Bitcraze AB, Malmö, Sweden) open-source platform and the Loco Positioning System (Bitcraze AB, Malmö, Sweden), which is based on the DecaWave DWM1000, were used for system implementation. The developed approach was implemented on the UAV MCU and tested in real-time closed-loop flight. Wang et al. [137] proposed an indoor RTLS for the localization of multiple UAVs using maximum likelihood Kalman filtering (MLKF). This work aimed to enhance the localization performance for flying objects in the vertical axis via the anchor's height level similarity. The work used Flame Wheel 450 (DJI Technology Inc., Shenzhen, China) UAVs with a wheelbase of 450 mm, equipped with an onboard Intel CPU (Intel Corporation, Santa Clara, CA, USA) for running the Ubuntu16 (Canonical Ltd., London, UK) operating system. Four UWB anchors were distributed in four corners of the testing environments and tested with several heights, from 90 cm to 180 cm. The used UWB system operated on channel 1, with 3.5 GHz central frequency and a 110 Kbps sample rate. High-precision motion-capture cameras were used for providing true position and speed. The practical system tests showed that the localization error was reduced by 60% by using the proposed MLKF, with only 20% of the computational time of the classical method. Zhang et al. [138] proposed a precise energy-saving indoor RTTS for UAVs in coal mines using an on-demand trigger algorithm based on an unscented Kalman filter. The filter fuses the data from UWB units and IMU units to achieve on-demand precise tracking (OPT). The P440 UWB module (TDSR LLC., Petersburg, TN, USA) was used to implement the anchors and tag. The Amovlab Z410-4b (Amu Lab-Chengdu Bobei Technology Co., Ltd., Chengdu, China) UAV was used for system implementation and testing, where the UWB tag was fixed on the UAV chassis. The practical tests were implemented in indoor laboratories as well as in coal mine indoor environments. The test results revealed that the proposed method improved the accuracy by 10.3% and 11.4% in indoor laboratory and coal mine environments, respectively, and the energy consumption was reduced by 19.6%. Zhang et al. [139] proposed an indoor UAV formation system based on a long short-term memory (LSTM) neural network. The LSTM network combined with raw data of UWB units was used for UAV positioning using historical information for enhancing the system performance. Five Firefighter UAVs (Yifei Intelligent Control Technology Co., Ltd., Tianjin, China) were used to hold the tags of the LinkTrack-P (SZ Nooploop Technology Co., Ltd., Shenzhen, China) positioning system, which also includes four anchors as a base station for practical system testing in indoor environments. The test results indicated that the system achieved a positioning accuracy of 5 cm. Lin et al. [140] proposed a UWB-incorporated visual-inertial odometry (VIO) system for UAV indoor navigation in GNSS-denied environments. Their work aimed to improve localization accuracy by combining the VIO with the ultra-wideband (UWB) positioning technology. A commercial UAV controlled by the Cube Pixhawk2 (CubePilot Pty. Ltd., Breakwater, Australia) and equipped with a 3DM-GX5-25 (LORD Corporation, Williston, VT, USA) IMU and the Intel RealSense D455 depth camera (Intel Corporation, Santa Clara, CA, USA) was used for practical system testing. The LinkTrack S (SZ Nooploop Technology Co., Ltd., Shenzhen, China) UWB-based IPS, which is based on the TDoA ranging approach, was used, including four fixed anchors and a movable tag. The experimental test results showed that the proposed method achieved an accuracy of 41.4 cm, which was better than the individual performance of UWB (46.3 cm) and VIO (1.506 m). Niculescu et al. [141] proposed an energy-efficient 3D indoor RTLS for UAVs. They used UWB ranging with wake-up radio (WUR) technologies to minimize the modules' power consumption during data exchange between the tags on the UAVs and the anchors. Their approach was composed of a lightweight positioning algorithm, optimal flight strategy, and ranging error correction algorithm. The system hardware consisted of the Crazyflie 2.1 (Bitcraze AB, Malmö, Sweden) UAV with the STM32F405 Cortex-M4 (STMicroelectronics, Geneva, Switzerland) microcontroller, which

enables the flight control, the positioning algorithm implementation, and the integration of BMI088 and BMP388 (Bosch Sensortec GmbH, Reutlingen, Germany) IMU units and pressure sensors, as well as the UWB tag for positioning and localization. A custom-designed UWB-Wakeup shield based on the Qorvo DW1000 transceiver was used for UWB anchors and tag implementation at a 6.8 Mbit/s sample rate. The UWB-Wakeup shield also had the CC1200 (Texas Instruments Incorporated, Dallas, TX, USA) radio transceiver for sending on-off-key wake-up signals to the WUR receivers at 868 MHz. The system testing results showed that the localization error was within 28 cm at 1.2 ms MCU processing time, and it was improved by an additional 25% with the error correction algorithm. The energy consumption was determined to be 24 mJ at the sensor node, which is 50 times less compared to a traditional system with the same localization accuracy. Other UAV-based IPSs were proposed in [142–148].

6. Results

The works selected for this review were investigated for several factors to ensure that they served the purpose of our work, which was to facilitate the study and selection of real-time indoor tracking and positioning systems operating with ultra-wideband technology. The first factor was suitability for the indoor environment. This point is essential because of the limited number of systems that operate efficiently in indoor environments, due to the difficulty of UWB wave transmission between IPS units without passing through the obstacles that are a feature of these places, such as walls, furniture, doors, windows, etc. The second factor that we emphasized was the possibility of using the selected works to track moving objects such as robots, drones, etc., as well as the ease of integrating them with these platforms. The third factor that we focused on was the ability of the selected works to display the location of the objects to be tracked, whether using user graphical interfaces, modeling software, or any programs that can be used to monitor moving objects in real time.

We summarized the topology and ranging technique, the used UWB chip/module, the used sensors (hybrid), the used UWB frequency, the optimization algorithm/filter, the application, the performance improvement, the system positioning accuracy, the monitoring program/software, and the work reference number and year of the included IPSs/RTLSSs, as shown in Table 3.

Table 3. Summary of selected IPSs.

Topology	UWB Chip/Module	Multi-Sensor	UWB Frequency/Channel	Optimization Algorithm/Filter	Application	Improvement (%) / Accuracy (cm)	Monitoring	Reference No., Year
SS-TWR	DW1000	UWB	5 GHz	KE, PF	IPS	13 cm	MATLAB	[79], 2023
DS-TWR	DW3000	UWB	6.5 GHz, CH5	Trilateration, LSM	IPS, Robotic	50 cm	Python	[80], 2024
SS-TWR	DW1000	UWB	3.5–6.5 GHz	Trilateration	IPS	14 cm	Custom	[81], 2020
DS-TWR	DW1000	UWB	3.5–6.5 GHz	KF	IPS	8%	Custom	[82], 2021
TWR	DW1000	UWB+IMU	3.5–6.5 GHz	DUKE, RMSE	IPS	6 cm	MATLAB	[83], 2024
AltDS-TWR, TDoA	DW-EVK1000	UWB	3.5–6.5 GHz	Multilateration	IPS	-	Custom	[84], 2020
SDS-TWR, AltDS-TWR	ElikoUWB-RTLS	UWB	3.5–6.5 GHz	Active-passive method	RTTS	20%	R Language	[85], 2022
TWR	Wi-Pos UWB	UWB	CH5, 6.5 GHz	KF	IPS	50%, 15 cm	Custom	[86], 2024
TDoA	DW1000	UWB	3.5–6.5 GHz	Trilateration	RTTS	9 cm	Web-based application	[98], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	Taylor algorithm	IPS	10 cm	Custom	[99], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	LSTM, DeepTAL	IPS	20 cm	Custom	[100], 2019
TWR, TDoA	DW3000	UWB	CH5, 6.5 GHz	EKF	IPS	38%, 22 cm	Custom	[103], 2023
TDoA	Custom	IR, UWB	7.25 GHz	Multilateration, median, mean F.	RTLS	10 cm	Custom	[104], 2021

Table 3. Cont.

Topology	UWB Chip/Module	Multi-Sensor	UWB Frequency/Channel	Optimization Algorithm/Filter	Application	Improvement (%) / Accuracy (cm)	Monitoring	Reference No., Year
TDoA	DW1000	UWB	3.5–6.5 GHz	-	IPS	15 cm	Custom	[105], 2020
PDoA	Custom	UWB, IMU, GPS	6.5 GHz	Chan and convex method	IPS	45 cm	Custom	[110,111], 2021
AoA, PDoA	DecaWave Beta-PDoA	UWB	3.5–6.5 GHz	Triangulation	IPS	-	KDE plots	[112], 2022
wrapped PDoA	DW1000	UWB	3.5–6.5 GHz	GMM	RTLS	8 cm	Custom	[113], 2022
TWR	DW1000	UWB, IMU, Camera	3.5–6.5 GHz	MCKF	Robotics, RTLS	13.7 cm	SLAM	[122], 2019
TWR	P440	UWB, IMU	4.3 GHz	ESKF	Robotics, RTLS	20 cm	Custom	[4], 2020
TWR	DW1000	UWB	3.5–6.5 GHz	M-RKF	IPS, robotics	31%, 30 cm	Custom	[123], 2023
TDoA	Pozyx UWB	UWB, LiDAR	CH5, 6.5 GHz	MVG, EKF, LPF	RTLS, robotics	6.16%, 8 cm	Python	[124], 2024
TWR	Custom	UWB, LiDAR	-	EKF, PDR	IPS, robotics	47.8 cm	Custom	[125], 2024
TWR	MDEK1001	UWB LiDAR	3.5–6.5 GHz	Trilateration	IPS, robotics	7 cm	Custom	[126], 2021
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	UKF	IPS, UAV	15 cm	Custom	[136], 2022
TWR	Custom	UWB	CH1, 3.5 GHz	MLKF	RTLS, UAV	60%, 22.4 cm	Custom	[137], 2021
TWR	P440	UWB, IMU	4.3 GHz	ET-EKF	IPS, UAV	10.3%,	Gazebo platform	[138], 2022
TDoA	LinkTrack-P	UWB, IMU	3.5–6.5 GHz	LSTM-RNN, EKF	IPS, UAV, robotics	5 cm	Custom	[139], 2023
TDoA	LinkTrack-S	UWB, IMU, camera	3.5–6.5 GHz	Custom	IPS, UAV, robotics	41.4 cm	SLAM	[140], 2023
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	Multilateration	RTLS, IPS, UAV	25%	Custom	[141], 2023
PDoA	DW3000	UWB	CH5, 6.5 GHz	Distance adaptive weighting	IPS	36 cm	Custom	[149], 2023
TDoA	LinkTrack	UWB, IMU, camera	3.5–6.5 GHz	-	IPS, UAV	5 cm	Custom	[143], 2020
TDoA	DW1000	UWB	3.5–6.5 GHz	CWCS	IPS, robotics	20 cm	Custom	[150], 2024
TDoA	DW1000	UWB, IMU	3.5–6.5 GHz	Trilateration, Kalman F.	IPS, RTLS	94.7%, 8.25 cm	Custom	[151], 2022
AoA, TDoA	DW1000	UWB, IMU	3.5–6.5 GHz	PAAL, UKF	RTLS	12%, 12.5 cm	Custom	[152], 2021
TWR, TDoA	DW1000	UWB	3.5–6.5 GHz	Spoofing detection	RTLS	30 cm	Custom	[153], 2024
TDoA	DW1000	UWB	4 GHz, 6 GHz	Aperture-coupled patch	IPS, RTLS, UAV	Size of an apple	Custom	[154], 2022
DS-TWR	MDEK1001	UWB	3.5–6.5 GHz	Trilateration	RTLS	23.76 cm	Custom	[155], 2021
TDoA	EVK1000	UWB	3.5–6.5 GHz	TDMA	RTLS, Robotic	<30 cm	Python	[156], 2019
AltDS-TWR	TREK1000	UWB	3.5–6.5 GHz	Multilateration, KF	RTLS	<28 cm	Custom	[157], 2019
SDS-TWR	MAX2000/DW1000	UWB, IMU	3.5–6.5 GHz	EKF	RTLS, IPS	38.24%, 11 cm	Custom	[91], 2021
DS-TWR, TDoA	DW1000	UWB	3.5–6.5 GHz	KF	IPS	10 cm	MATLAB	[90], 2022
SS-TWR	DW3000	UWB	CH5, 6.5 GHz	Trilateration, WAKF	IPS	20.84%, 25 cm	Custom	[158], 2023
EDS-TWR	DW1000	UWB	3.5–6.5 GHz	EKF	IPS	80.53%, 51.8 cm	Custom	[92], 2024

Table 3. Cont.

Topology	UWB Chip/Module	Multi-Sensor	UWB Frequency/Channel	Optimization Algorithm/Filter	Application	Improvement (%) / Accuracy (cm)	Monitoring	Reference No., Year
TWR-TDoA	DW1000	UWB	3.5–6.5 GHz	MPGA	RTLS	11.21 cm	Python	[159], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	ML	IPS	29%, 46.6 cm	Custom	[160], 2021
TDoA	DW1000	UWB	3.5–6.5 GHz	MCC-VC	IPS	42 cm	MATLAB	[161], 2023
TDoA	Trimension NXP SR040	UWB, Bluetooth	6–8.5 GHz	Trilateration, LS	IPS	35 cm	Samsung's app	[101,102], 2024
TDoA	DW1000	UWB	3.5–6.5 GHz	WIP, DNN, LSTM	IPS	8.1 cm	Custom	[162], 2021
TDoA	DW1000	UWB	3.5–6.5 GHz	-	IPS	26 cm	Custom	[163], 2019
TDoA	DW1000	UWB	CH5, 6.5 GHz	EKF	IPS	<50 cm	Custom	[164], 2020
TDoA	DW1000	UWB, IMU	3.5–6.5 GHz	UKF	IPS	34 cm	Custom	[165], 2019
TDoA	MDEK1001	UWB	3.5–6.5 GHz	LS, KF	IPS	15 cm	SLAM	[97], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	LinHPS	IPS	10 cm	Custom	[166], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	PZS, DBLD	IPS	≈27 cm	SLAM	[17], 2021
TDoA, FDoA	DW1000	UWB	3.5–6.5 GHz	WLS, KF	IPS	≈60 cm	Custom	[107], 2021
SS-TWR, TDoA, FDoA	Custom	UWB	CH5, 6.5 GHz	BYOPS	IPS, robotics	40 cm	UCAS	[167], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	-	IPS, robotics	30 cm	Custom	[108], 2021
TDoA	DW1000	UWB	3.5–6.5 GHz	Trilateration	IPS	10 cm	Python	[168], 2023
TDoA	NXP-SR100T,	UWB	6–8.5 GHz, CH5, CH9	KF, RNN	IPS	31%	Custom	[169], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	EKF	RTLS	17 cm	C language	[109], 2019
TDoA	DW1000	UWB	3.5–6.5 GHz	SWF	IPS	20 cm	Custom	[170], 2020
TDoA	DW1000	UWB	3.5–6.5 GHz	FLA	IPS	17.6 cm	Custom	[171], 2019
TDoA	DW1000	UWB	3.5–6.5 GHz	LS	IPS	8 cm	Custom	[172], 2024
TDoA	DW1000	UWB	3.5–6.5 GHz	EKF	RTLS	19 cm	Custom	[173], 2019
TDoA	DW1000	UWB, IMU	3.5–6.5 GHz	Dead Reckoning, KF	IPS, robotics	10 cm	Custom	[174], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	ARPE,	IPS	50%	Custom	[175], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	Triangulation	IPS	9 cm	Custom	[176], 2022
TWR, TDoA	DW1000	UWB	3.5–6.5 GHz	-	IPS	30 cm	Custom	[177], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	EKF	IPS	30 cm	Custom	[178], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	Multilateration, DF, EKF	IPS	15.8 cm	Custom	[179], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	EKF	IPS	30.3 cm	Custom	[180], 2023
TWR	DW1000	UWB	3.5–6.5 GHz	MAL	IPS	20 cm	Custom	[181], 2022
TWR	DW1000	UWB, GPS	3.5–6.5 GHz	Trilateration	IPS	11.8 cm	Custom	[182], 2021
TDoA	DW1000	UWB	3.5–6.5 GHz	IGN	RTLS	50 cm	Custom	[183], 2020

Table 3. Cont.

Topology	UWB Chip/Module	Multi-Sensor	UWB Frequency/Channel	Optimization Algorithm/Filter	Application	Improvement (%) / Accuracy (cm)	Monitoring	Reference No., Year
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	UKF	IPS	50.28%, 48.7 cm	Custom	[184], 2024
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	Trilateration, EKF	RTLS	31 cm	Unity software	[185], 2023
TWR	DW1000	UWB, LiDAR	3.5–6.5 GHz	VGICP	IPS, robotics	40 cm	SLAM	[186], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	MCA	IPS, robotics	30 cm	Custom	[187], 2021
TWR	P440	UWB, IMU	4.3 GHz	REKF	IPS	56 cm	Custom	[188], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	EKF	IPS	60.61%, 10 cm	Custom	[189], 2020
TDoA	DW1000	UWB	3.5–6.5 GHz	EKF	IPS, robotics, UAV	42.09%, 14.1 cm	Custom	[190], 2021
TWR	DW1000	UWB	3.5–6.5 GHz	KF	IPS, robotics	16 cm	Custom	[191], 2024
TWR	Pozyx UWB	UWB	3.5–6.5 GHz	EKF	RTLS, robotics	20 cm	ROS	[192], 2020
TWR	Mini 3s	UWB, LiDAR	3.5–6.5 GHz	EKF	IPS, robotics	<40 cm	SLAM-ROS	[193], 2019
DS-TWR	DW1000	UWB, IMU	3.5–6.5 GHz	KF	IPS	7.58 cm	Custom	[194], 2020
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	EKF	IPS, robotics, UAV	60%, 48 cm	Custom	[195], 2020
TWR	Samsung UM100	UWB, camera	3.1–8.976 GHz	EKF	IPS	20 cm	SLAM	[196], 2020
TWR	Pozyx UWB	UWB, encoder	3.5–6.5 GHz	EKF	RTLS, robotics	6 cm	Custom	[135], 2023
TWR	Pozyx UWB	UWB, LiDAR, IMU	3.5–6.5 GHz	EKF	RTLS, robotics	10 cm	SLAM-ROS	[128], 2020
TWR	DW1000	UWB	3.5–6.5 GHz	CRLB	IPS, robotics	12 cm	Custom	[129], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	EKF	IPS, robotics	54 cm	Custom	[133], 2020
TDoA	DW1000	UWB	3.5–6.5 GHz	NN	IPS, robotics	89.2 cm	Custom	[197], 2024
TWR	DW1000	UWB	3.5–6.5 GHz	EFIR-RTS	IPS, robotics	25.35%, 16 cm	SLAM	[198], 2022
TWR	DW1000	UWB, LiDAR	3.5–6.5 GHz	UKF	IPS, robotics, UAV	34 cm	SLAM	[199], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	-	IPS, robotics, UAV	34.6 cm	Custom	[142], 2021
TWR	DW1000	UWB	3.5–6.5 GHz	Multilateration	IPS, UAV	28 cm	Custom	[141], 2023
TWR	Pozyx UWB	UWB, LiDAR, IMU	3.5–6.5 GHz	EKF	IPS, UAV	39 cm	SLAM	[200], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	EKF	IPS, UAV	32%, 6.9 cm	Custom	[146], 2024
SS-TWR	DW1000	UWB	3.5–6.5 GHz	Multilateration, GDoP	IPS, robotics, UAV	15 cm	SLAM	[201], 2022
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	EKF	RTLS, UAV	15 cm	Custom	[202], 2023
TDoA	LinkTrack-S	UWB, IMU, camera	3.5–6.5 GHz	UWB odometry-constraint	IPS, UAV	43.4 cm	SLAM	[203], 2022
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	EKF	IPS, UAV	20.8 cm	Custom	[145], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	EKF	IPS, UAV	27.8 cm	Custom	[204], 2023
TWR	DW1000	UWB	3.5–6.5 GHz	Multilateration	IPS, UAV	10 cm	Custom	[144], 2020

Table 3. Cont.

Topology	UWB Chip/Module	Multi-Sensor	UWB Frequency/Channel	Optimization Algorithm/Filter	Application	Improvement (%) / Accuracy (cm)	Monitoring	Reference No., Year
TDoA	LinkTrack-S	UWB, IMU	3.5–6.5 GHz	EKF	IPS, UAV	3.8 cm	Custom	[148], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	STVBF	IPS, UAV	42.0%, 40 cm	Custom	[205], 2022
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	Trilateration, KF	IPS, UAV	10 cm	C++	[206], 2021
TWR	DW1000	UWB, IMU, camera	3.5–6.5 GHz	Outlier rejection algorithm	IPS, UAV	20%, 14 cm	SLAM	[207], 2021
TDoA	LinkTrack-S	UWB, IMU, camera	3.5–6.5 GHz	CNN, LSTM	IPS, UAV	42 cm	Custom	[208], 2023
TWR	Pozyx UWB	UWB	3.5–6.5 GHz	GDoP	IPS, UAV	20 cm	Custom	[209], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	Multilateration	IPS, UAV	30 cm	Custom	[210], 2023
TWR	DW1000	UWB	3.5–6.5 GHz	ANN	RTLS	26 cm	Custom	[211], 2024
TWR	DW1000	UWB	3.5–6.5 GHz	LS	IPS, robotics	20 cm	Custom	[212], 2023
TDoA	DW1000	UWB	3.5–6.5 GHz	DoP	IPS	44 cm	Custom	[213], 2022
TDoA	LinkTrack	UWB, LiDAR	3.5–6.5 GHz	PF	IPS, robotics	31.2 cm	Custom	[131], 2021
TDoA	DW1000	UWB	3.5–6.5 GHz	PDoP	IPS	20 cm	Custom	[214], 2021
DS-TWR	DW1000, DW3000	UWB	3.5–6.5 GHz	Multilateration	IPS	40 cm	MATLAB	[215], 2024
SDS-TWR	DW1000	UWB	3.5–6.5 GHz	Trilateration	RTLS	10 cm	Custom	[216], 2023
DS-TWR	DW1000	UWB	3.5–6.5 GHz	EKF, LSTM	IPS	50%, 37 cm	MATLAB	[217], 2023
AoA, TDoA	DW1000, Ubisense D4	UWB	3.5–6.5 GHz	EKF, PF	RTLS	40 cm	Custom	[218], 2020
TWR	DW1000	UWB	3.5–6.5 GHz	Multilateration	RTLS	53 cm	Python	[219], 2023
TWR	DW1000	UWB LiDAR	3.5–6.5 GHz	Trilateration, LS	IPS, robotics	16.34 cm	Python	[220], 2024
TWR	DW1000	UWB	3.5–6.5 GHz	LS, ECA	IPS	17 cm	Python	[221], 2023
TWR	LinTrack	UWB, IMU	3.5–6.5 GHz	EKF	IPS, UAV	51 cm	Custom	[222], 2021
TWR	DW1000	UWB	3.5–6.5 GHz	ANN	IPS	88.45%	Custom	[76], 2022
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	PF, EKF	IPS	36.15%, 71.5 cm	MATLAB	[223], 2024
TWR	DW1000	UWB	3.5–6.5 GHz	CNN	IPS	7.35 cm	Custom	[224], 2024
TWR	DW1000	UWB LiDAR	3.5–6.5 GHz	EKF, PF	IPS, robotics	7.2 cm	SLAM	[225], 2023
TWR	DW1000	UWB	3.5–6.5 GHz	Trilateration, KNN	IPS	4.71 cm	Custom	[226], 2021
TWR	DW1000	UWB	3.5–6.5 GHz	Trilateration, UKF	IPS	4 cm	Custom	[227], 2024
TWR	DW1000	UWB, Wi-Fi	CH12, 3.993 GHz	KNN	IPS	69.8%, 30 cm	Custom	[228], 2024
TWR	Pozyx UWB	UWB	3.5–6.5 GHz	Trilateration	IPS	30 cm	Python	[229], 2024
TDoA	DW1000	UWB	3.5–6.5 GHz	Triangulation, KF	IPS	68%, 10 cm	Web/mobile application	[230], 2023
TWR	Pozyx UWB	UWB	3.5–6.5 GHz	ANN, KNN	IPS	34%, 31.6 cm	Python	[231], 2022
ADS-TWR	DW1000	UWB	3.5–6.5 GHz	Multilateration, EKF	IPS	-	Python	[232], 2023

Table 3. Cont.

Topology	UWB Chip/Module	Multi-Sensor	UWB Frequency/Channel	Optimization Algorithm/Filter	Application	Improvement (%) / Accuracy (cm)	Monitoring	Reference No., Year
TWR	Pozyx UWB	UWB, Wi-Fi	3.5–6.5 GHz	Trilateration	IPS	87 cm	Python	[233], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	SMA	IPS	6.69 cm	Custom	[234], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	Trilateration, KF	IPS, robotics	38%, 12.4 cm	Custom	[235], 2022
TWR	DW1000	UWB	3.5–6.5 GHz	WHFFA	IPS	4 cm	Custom	[236], 2023
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	FNN	IPS	25 cm	Custom	[237], 2023
DS-TWR	DW1000	UWB, IMU	3.5–6.5 GHz	Multilateration, LS, EKF	IPS	20 cm	Custom	[238], 2022
TWR	DW1000	UWB, IMU	3.5–6.5 GHz	Trilateration	IPS	20 cm	Custom	[239], 2022
TDoA	DW1000	UWB	3.5–6.5 GHz	Trilateration	IPS	33.4 cm	Custom	[240], 2019
TWR	DW1000	UWB	3.5–6.5 GHz	LS, Chan	IPS	57.6%, 15.7 cm	Python	[241], 2022
AoA, PDoA	DW1000	UWB, IMU	3.5–6.5 GHz	Triangulation	IPS	10 cm	Custom	[242], 2021

7. Discussion

The results listed in Table 3 provide a clear picture of the prevailing trend in the implementation of indoor tracking and positioning systems. The data provide important information about the most popular available UWB hardware components, modules, and units that can be relied upon to implement the required IPSs in real time. The dominance of the products of Qorvo Inc. (DecaWave) in the market of this field is clearly evident, as the percentage of works that relied on direct components from this company reached approximately 73.5% (108/147) of the systems included in the current work—especially the popular transceiver DW1000, which is the core of 101 out of 108 presented IPSs. There are also some systems and products with private brands that rely on the use of DW1000 units in their structure. This spread is due to several factors, including the scarcity of similar alternatives, its low price, and its high accuracy, resulting in the development of many commercial UWB systems. The new and improved generation DW3000 has been launched, which has many improvements, especially in the field of energy consumption, where it consumes 50% less energy than DW1000 [243]. It also supports higher SPI bus speed and can work on UWB channel 9, making it a better choice for new designs. Figure 13 explains the distribution of the used UWB modules in this study.

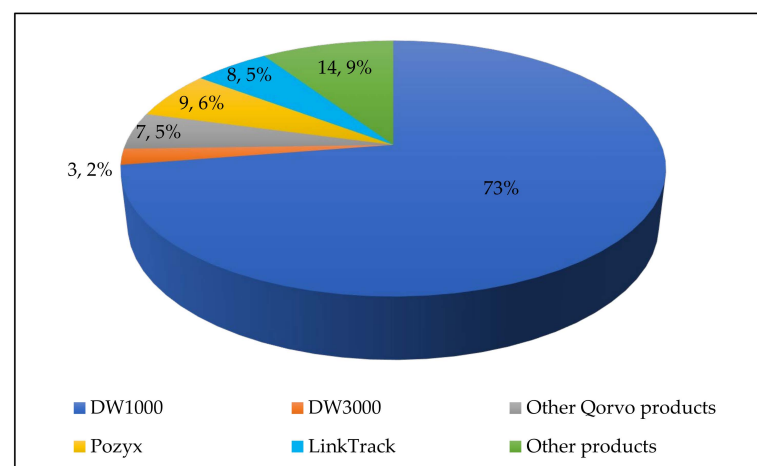


Figure 13. Distribution of the used UWB modules.

Several topologies have been used to measure the distances between the IPS UWB components to allow the positioning and localization algorithm to calculate the exact location and provide the coordinates of the tracked movable object. In the present work, the TWR method had the highest implementation in IPSs, at 78/147 (see Figure 14). This is the result of this method's simplicity of implementation, without the need for anchors to synchronize complex hardware and software requirements. The TDoA approach was used in 52.35% of the included systems, while the PDoA and hybrid implementations accounted for 6.4% and 11.8%, respectively.

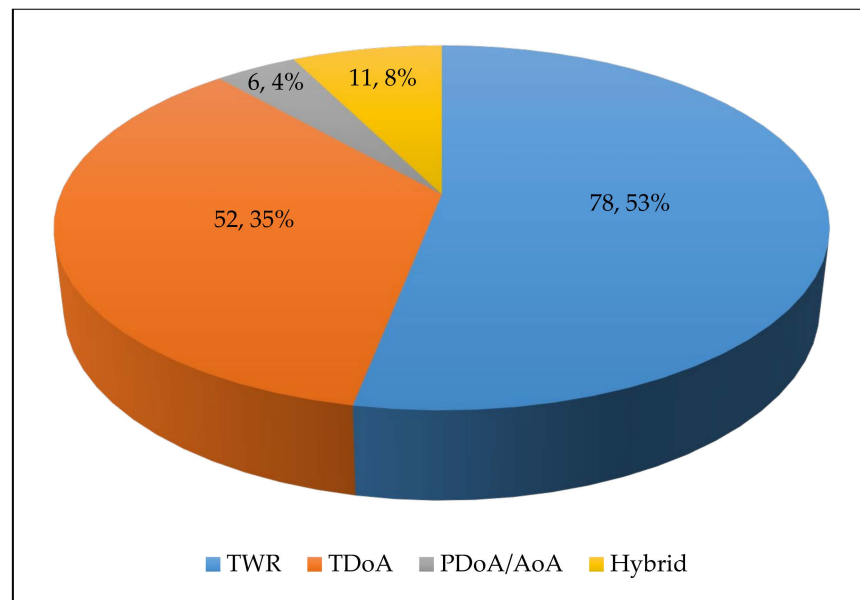


Figure 14. Distribution of the used UWB topologies.

Some IPSs use other sensors besides the UWB technology to enhance the system performance using sensor fusion algorithms such as Kalman filters and ANNs. In the present work, 30% of the proposed IPSs used different types of sensors with UWB, such as IMUs, vision sensors/cameras, encoders, LiDAR/IR, acoustic/ultrasound, GPS, Bluetooth, and Wi-Fi (see Figure 15).

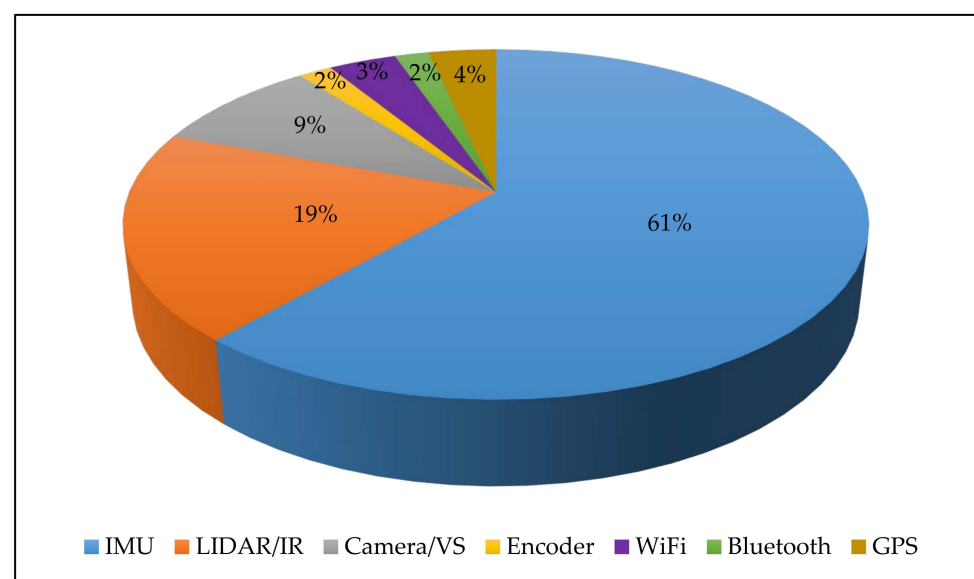


Figure 15. Distribution of the sensors and supporting technologies besides the UWB.

Many algorithms and filters have been used to reduce position errors caused by measurement errors in ultra-wideband technology, especially when passing through obstacles in non-line-of-sight environments. These algorithms and filters are also used to fuse data from more than one source when using indoor positioning systems operating with different types of technologies and sensors. Figure 16 explains the distribution of the used algorithms and filters in the proposed systems.

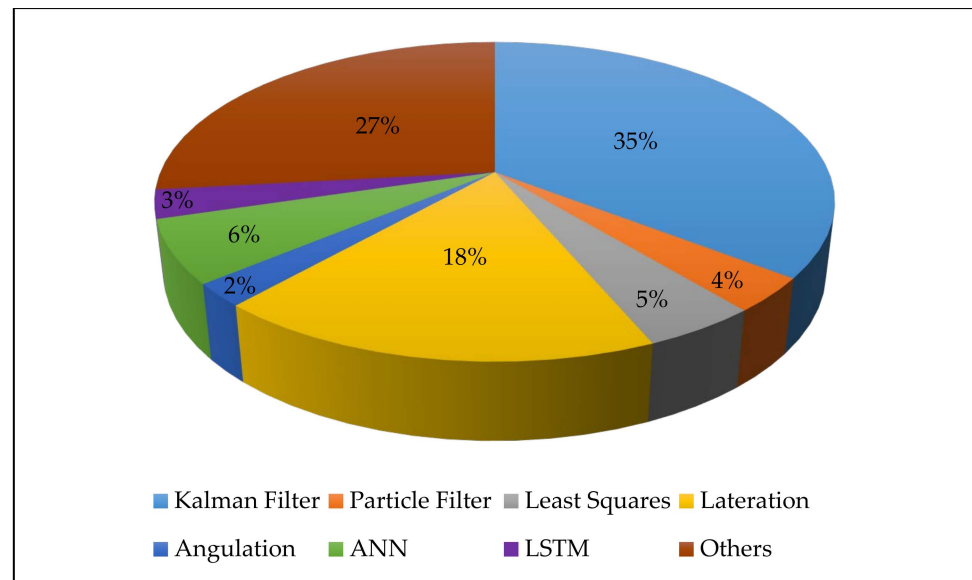


Figure 16. Distribution of the used algorithms and filters.

Indoor positioning and tracking systems can be used in many applications, whether for personal uses such as tracking and monitoring systems for the disabled and the elderly in the health field, or to track the movement of valuable private property. They can also be used for industrial applications, resource management, and in warehouses, e.g., for tracking the movement of goods, forklifts, robots, and unmanned ground and air vehicles. Figure 17 explains the most important applications for the included references.

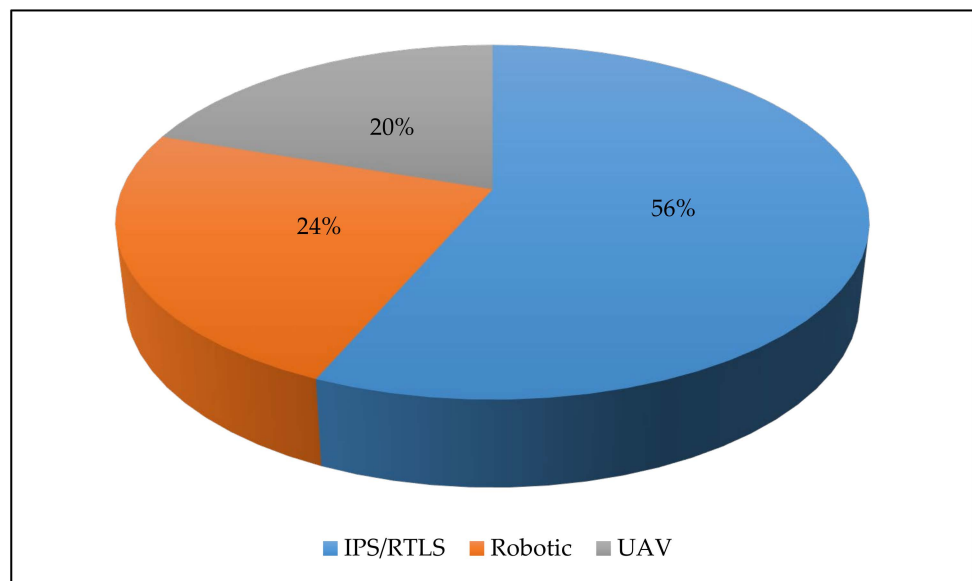


Figure 17. The main applications of the proposed references.

The accuracy data in Table 3 show that the accuracy varies between the different systems. This is logical because the testing environments and conditions differ in terms of

the numbers of anchors and tags used, as well as the types of algorithms used to calculate the distances between the system units or to reduce and filter errors during the identification and tracking of moving objects. The largest error rate recorded was 89.2 cm. Figure 18 shows the percentage range of errors recorded for the systems included in this study.

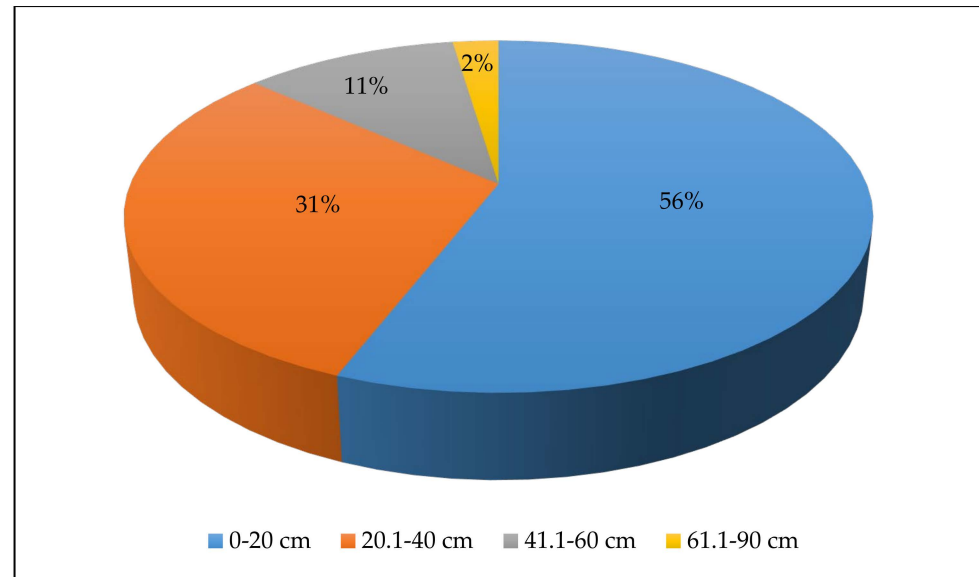


Figure 18. The percentage range of errors for the included systems.

Despite the importance of monitoring programs that convert ultra-wideband unit measurements into distances and plot the coordinates and locations of moving objects, many works lack information on how this part was implemented and which software or programming languages were used. The use of MATLAB was noted in 7/147 works, while Python was used in 12/147 works. Robot tracking systems and SLAM systems were used in 16 of the works presented here, while various programs were used in 6/147 works. The type of program or language used was not mentioned in the remaining works.

8. Conclusions

Ultra-wideband technology offers attractive features such as its accuracy, robustness, and ability to penetrate obstacles, making it one of the preferred choices for implementing real-time indoor positioning and tracking systems. In this review, we investigated the real-time indoor positioning, tracking, and localization systems developed during the past 5 years. A total of 147 research papers were included this review of real-time positioning systems using different modules, units, and chips from different vendors, with systems implemented using different methods and algorithms. Basic information was extracted from the works included in the survey period, such as the type of hardware used and its origin, methods of implementing the systems, types of sensors and technology used, algorithms and filters to improve performance, and the accuracy of the measurements obtained. This review ensures a clear picture of the types of methods and algorithms used to implement these systems in real time. The selected works were addressed and discussed from several perspectives in terms of the hardware components used, the frequency range and ultra-wideband channels, the types of topologies implemented, the filters and algorithms to improve the performance, and the accuracy of the obtained performance, which we believe are important for any researcher in this field to build an initial idea about how to choose the appropriate type and method to implement this process in real time.

Author Contributions: Conceptualization, M.F.R.A.-O., T.R. and K.T.; formal analysis, M.F.R.A.-O. and K.T.; writing—original draft preparation, M.F.R.A.-O.; methodology, M.F.R.A.-O., S.J. and K.T.; writing—review and editing, M.F.R.A.-O. and K.T.; visualization, M.F.R.A.-O.; supervision, K.T.;

project administration, T.R. and K.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research is part of the Synergy Project ADAM (Autonomous Discovery of Advanced Materials), funded by the European Research Council (grant number: 856405).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Acknowledgments: The authors would like to thank the European Research Council (ERC) for funding the Autonomous Discovery of Advanced Materials (ADAM) project.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

ARPF	Adaptive robust particle filtering
ANN	Artificial neural network
BLE	Bluetooth Low Energy
CRLB	Cramér–Rao lower bounds
CNN	Convolutional neural network
DF	Debiasing filter
ECA	Efficient channel attention
FNN	Fuzzy neural network
GPS	Global Positioning System
BYOPS	Bring Your Own Positioning System
AltDS-TWR	Alternative double-sided two-way ranging
SDS-TWR	Symmetric double-sided two-way ranging
EDS-TWR	Extended double-sided two-way ranging
EFIR-RTS	Extended finite-impulse response-based Rauch–Tung–Striebel smoother
DS-TWR	Double-sided two-way ranging
DBLD	Dynamic best link discovery
DNN	Deep neural network
FLA	Five-line approximate
IPS	Indoor positioning system
ISAR	Inverse synthetic-aperture radar
IGN	Iterative Gauss–Newton
RF	Radio frequency
KF	Kalman filter
CKF	Cubature Kalman filter
KNN	k-Nearest neighbors
LoS	Line-of-sight
LS	Least squares
LSTM	Long short-term memory
LinHPS	Linear hyperbolic positioning system
ML	Machine learning
MAL	Multi-algorithm
MPGA	Multi-population genetic algorithm
MCC-VC	Maximum correntropy criterion with variable center
MCA	Multipath component analysis
NLoS	Non-line-of-sight
NN	Neural network
PF	Particle filter
REKF	Robust extended Kalman filter
RFID	Radio frequency identification
RSSI	Received signal strength indicator
RNN	Recurrent neural network
SAR	Synthetic-aperture radar
SWF	Sliding window filtering

STVBF-DO	Skew-t variational Bayes filter-disturbance observer
TDoA	Time difference of arrival
ToF	Time of flight
TWR	Two-way ranging
UWB	Ultra-wideband
UAV	Unmanned aerial vehicle
PAAL	Passive anchor-assisted localization
PDoA	Phase difference of arrival
PZS	Predictive zone selection
PDoP	Position dilution of precision
GDoP	Geometric dilution of precision
CWCS	Cascaded wireless clock synchronization
KF	Kalman filter
ROS	Robot operating system
WAKF	Weighted adaptive Kalman filter
WIP	Weighted indoor positioning
UKF	Unscented Kalman filter
UCAS	UWB control and analysis suite
EKF	Extended Kalman filter
TDMA	Time-division multiple access
TDoA	Time difference of arrival
SMA	Slime mold algorithm
VGICP	Voxelized generalized iterative closest point
WHFFA	Weighted hybrid filter following algorithm

References

- Indoor Positioning: Technologies and Performance | IEEE eBooks | IEEE Xplore. Available online: <https://ieeexplore.ieee.org/book/8786839> (accessed on 26 March 2024).
- Lee, C.K.M.; Ip, C.M.; Park, T.; Chung, S.Y. A Bluetooth Location-based Indoor Positioning System for Asset Tracking in Warehouse. In Proceedings of the 2019 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Macao, China, 15–18 December 2019; pp. 1408–1412.
- Chan, P.Y.; Chao, J.-C.; Wu, R.-B. A Wi-Fi-Based Passive Indoor Positioning System via Entropy-Enhanced Deployment of Wi-Fi Sniffers. *Sensors* **2023**, *23*, 1376. [CrossRef] [PubMed]
- Li, M.-G.; Zhu, H.; You, S.-Z.; Tang, C.-Q. UWB-Based Localization System Aided With Inertial Sensor for Underground Coal Mine Applications. *IEEE Sens. J.* **2020**, *20*, 6652–6669. [CrossRef]
- Kloock, M.M.; Scheffe, P.; Maczjewski, J.; Alrifae, B.; Kowalewski, S.; Tülleners, I. *Vision-Based Real-Time Indoor Positioning System for Multiple Vehicles*; Elsevier: Amsterdam, The Netherlands, 2020; Volume 53, p. 15446.
- Höflinger, F.; Hoppe, J.; Zhang, R.; Ens, A.; Reindl, L.; Wendeborg, J.; Schindelhauer, C. Acoustic indoor-localization system for smart phones. In Proceedings of the 2014 IEEE 11th International Multi-Conference on Systems, Signals & Devices (SSD14), Castelldefels-Barcelona, Spain, 11–14 February 2014; pp. 1–4.
- Khan, S.I.; Ray, B.R.; Karmakar, N.C. RFID localization in construction with IoT and security integration. *Autom. Constr.* **2024**, *159*, 105249. [CrossRef]
- White Papers—Qorvo. Available online: <https://www.qorvo.com/resources/d/qorvo-getting-back-to-basics-with-ultra-wideband-uw-b-white-paper> (accessed on 11 July 2024).
- Alarifi, A.; Al-Salman, A.; Alsaleh, M.; Alnafessah, A.; Al-Hadhrani, S.; Al-Ammar, M.A.; Al-Khalifa, H.S. Ultra Wideband Indoor Positioning Technologies: Analysis and Recent Advances. *Sensors* **2016**, *16*, 707. [CrossRef]
- Pierlot, V.; Van Droogenbroeck, M. A New Three Object Triangulation Algorithm for Mobile Robot Positioning. *IEEE Trans. Robot.* **2014**, *30*, 566–577. [CrossRef]
- Indoor Positioning Based on Time of Flight and Kalman Filtering Using Ultra-Wideband Sensors. Available online: <https://ieeexplore.ieee.org/document/10488216> (accessed on 19 June 2024).
- Asaad, S.M.; Maghddid, H.S. A Comprehensive Review of Indoor/Outdoor Localization Solutions in IoT era: Research Challenges and Future Perspectives. *Comput. Netw.* **2022**, *212*, 109041. [CrossRef]
- Monica, S.; Ferrari, G. UWB-based localization in large indoor scenarios: Optimized placement of anchor nodes. *IEEE Trans. Aerosp. Electron. Syst.* **2015**, *51*, 987–999. [CrossRef]
- Sundin, O. UWB-TWR Performance Comparison in a Hybrid Node Network. Master’s Thesis, Luleå University of Technology, Luleå, Sweden, 2021. Available online: <https://ltu.diva-portal.org/smash/record.jsf?pid=diva2:1602910&dswid=-59> (accessed on 25 June 2024).
- Zhao, K.; Zhao, T.; Zheng, Z.; Yu, C.; Ma, D.; Rabie, K.; Kharel, R. Optimization of Time Synchronization and Algorithms with TDOA Based Indoor Positioning Technique for Internet of Things. *Sensors* **2020**, *20*, 6513. [CrossRef]

16. Synchronization of Anchors in a TDoA Based Ultra-Wide Band Localization System. Available online: <https://research.tue.nl/en/studentTheses/synchronization-of-anchors-in-a-tdoa-based-ultra-wide-band-locali> (accessed on 4 April 2024).
17. Friedrich, J.; Tiemann, J.; Wietfeld, C. Accurate Multi-Zone UWB TDOA Localization utilizing Cascaded Wireless Clock Synchronization. In Proceedings of the 2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN), Lloret de Mar, Spain, 29 November–2 December 2021; pp. 1–8.
18. Djaja-Josko, V.; Kolakowski, J. A new method for wireless synchronization and TDOA error reduction in UWB positioning system. In Proceedings of the 2016 21st International Conference on Microwave, Radar and Wireless Communications (MIKON), Krakow, Poland, 9–11 May 2016; pp. 1–4.
19. Chen, H.; Ballal, T.; Saeed, N.; Alouini, M.-S.; Al-Naffouri, T.Y. A Joint TDOA-PDOA Localization Approach Using Particle Swarm Optimization. *IEEE Wirel. Commun. Lett.* **2020**, *9*, 1240–1244. [\[CrossRef\]](#)
20. Wolf, F.; Berg, V.; Dehmas, F.; Mannoni, V.; De Rivaz, S. Multi-Frequency Phase Difference of Arrival for Precise Localization in Narrowband LPWA Networks. In Proceedings of the ICC 2021-IEEE International Conference on Communications, Montreal, QC, Canada, 14–23 June 2021; pp. 1–6.
21. Li, J.; Lv, S.; Lv, L.; Wu, S.; Liu, Y.; Nie, J.; Jin, Y.; Wang, C. Joint TDOA, FDOA and PDOA Localization Approaches and Performance Analysis. *Remote Sens.* **2023**, *15*, 915. [\[CrossRef\]](#)
22. Margiani, T.; Cortesi, S.; Keller, M.; Vogt, C.; Polonelli, T.; Magno, M. Angle of Arrival and Centimeter Distance Estimation on a Smart UWB Sensor Node. *IEEE Trans. Instrum. Meas.* **2023**, *72*, 9508110. [\[CrossRef\]](#)
23. Shah, S.; Chaiwong, K.; Kovavisaruch, L.-O.; Kaemarungsi, K.; Demeechai, T. Antenna Delay Calibration of UWB Nodes. *IEEE Access* **2021**, *9*, 63294–63305. [\[CrossRef\]](#)
24. Elsanhoury, M.; Mäkelä, P.; Koljonen, J.; Välisuo, P.; Shamsuzzoha, A.; Mantere, T.; Elmusrati, M.; Kuusniemi, H. Precision Positioning for Smart Logistics Using Ultra-Wideband Technology-Based Indoor Navigation: A Review. *IEEE Access* **2022**, *10*, 44413–44445. [\[CrossRef\]](#)
25. Wan, E.A.; Van Der Merwe, R. The unscented Kalman filter for nonlinear estimation. In Proceedings of the IEEE 2000 Adaptive Systems for Signal Processing, Communications, and Control Symposium (Cat. No.00EX373), Lake Louise, AB, Canada, 4 October 2000; pp. 153–158.
26. Yang, C.; Shi, W.; Chen, W. Comparison of Unscented and Extended Kalman Filters with Application in Vehicle Navigation. *J. Navig.* **2017**, *70*, 411–431. [\[CrossRef\]](#)
27. Elfring, J.; Torta, E.; van de Molengraft, R. Particle Filters: A Hands-On Tutorial. *Sensors* **2021**, *21*, 438. [\[CrossRef\]](#)
28. Navigine All about Particle Filter for Indoor Navigation and Positioning. Available online: <https://navigine.com/blog/particle-filter/> (accessed on 2 October 2024).
29. An Overview of DecaWave’s DW1000 UWB Wireless Transceiver for Precise Indoor Positioning. Available online: <https://www.symmetryelectronics.com/blog/an-overview-of-decawave-s-dw1000-uwb-wireless-transceiver/> (accessed on 15 July 2024).
30. DW1000—Qorvo. Available online: <https://www.qorvo.com/products/p/https://www.qorvo.com/products/p/DW1000> (accessed on 15 July 2024).
31. Pelka, M.; Goronzy, G.; Hellbrück, H. Iterative approach for anchor configuration of positioning systems. *ICT Express* **2016**, *2*, 1–4. [\[CrossRef\]](#)
32. DWM3000—Qorvo. Available online: <https://www.qorvo.com/products/p/https://www.qorvo.com/products/p/DWM3000> (accessed on 23 April 2024).
33. *IEEE Std 802.15.4-2011 Revis. IEEE Std 802154-2006*; IEEE Standard for Local and Metropolitan Area Networks—Part 15.4: Low-Rate Wireless Personal Area Networks (LR-WPANs). IEEE: New York, NY, USA, 2011; pp. 1–314. [\[CrossRef\]](#)
34. *IEEE Std 802.15.4z-2020 Amend. IEEE Std 802154-2020*; IEEE Standard for Low-Rate Wireless Networks—Amendment 1: Enhanced Ultra Wideband (UWB) Physical Layers (PHYs) and Associated Ranging Techniques. IEEE: New York, NY, USA, 2020; pp. 1–174. [\[CrossRef\]](#)
35. Trimension SR150. Available online: <https://www.nxp.com/products/wireless-connectivity/trimension-uwb/trimension-sr150-secure-uwb-solution-for-iot-devices:SR150> (accessed on 15 July 2024).
36. Ultra-Wideband (UWB). Available online: <https://www.nxp.com/applications/enabling-technologies/connectivity/ultra-wideband-uwb:UWB> (accessed on 15 July 2024).
37. Ultra-Wideband Solutions | Microchip Technology. Available online: <https://www.microchip.com/en-us/products/wireless-connectivity/specialty/ultra-wideband> (accessed on 15 July 2024).
38. *IEEE Std 802.15.4f-2012 Amend. IEEE Std 802154-2011*; IEEE Standard for Local and Metropolitan Area Networks—Part 15.4: Low-Rate Wireless Personal Area Networks (LR-WPANs) Amendment 2: Active Radio Frequency Identification (RFID) System Physical Layer (PHY). IEEE: New York, NY, USA, 2012; pp. 1–72. [\[CrossRef\]](#)
39. Infineon Technologies Switzerland AG—Technology. Available online: <https://www.3db-access.com/technology> (accessed on 15 July 2024).
40. Home. Available online: <https://www.sparkmicro.com/> (accessed on 16 July 2024).
41. B-UWB-MOD1—Ultra-Wideband Module for High-Precision Indoor Location—STMicroelectronics. Available online: https://www.st.com/en/wireless-connectivity/b-uwb-mod1.html?ecmp=tt9470_gl_link_feb2019&rt=db&id=DB4404#documentation (accessed on 2 October 2024).

42. DWM1000—Qorvo. Available online: <https://www.qorvo.com/products/p/https://www.qorvo.com/products/p/DWM1000> (accessed on 12 November 2024).
43. DW3110—Qorvo. Available online: <https://www.qorvo.com/products/p/https://www.qorvo.com/products/p/DW3110> (accessed on 12 November 2024).
44. DW3120—Qorvo. Available online: <https://www.qorvo.com/products/p/https://www.qorvo.com/products/p/DW3120> (accessed on 12 November 2024).
45. DW3220—Qorvo. Available online: <https://www.qorvo.com/products/p/https://www.qorvo.com/products/p/DW3220> (accessed on 12 November 2024).
46. Trimension SR040. Available online: <https://www.nxp.com/products/wireless/secure-ultra-wideband-uwband/trimension-sr040-secure-uwband-solution-for-iot-tags:SR040> (accessed on 12 November 2024).
47. ATA8350. Available online: <https://www.microchip.com/en-us/product/ATA8350> (accessed on 15 July 2024).
48. ATA8352. Available online: <https://www.microchip.com/en-us/product/ata8352> (accessed on 12 November 2024).
49. SR1010 & SR1020 TRANSCEIVERS. SPARK Microsystems. Available online: <https://www.sparkmicro.com/product/sr1010-sr1020-transceivers/> (accessed on 12 November 2024).
50. Inpixon Inpixon: Real-Time Location Systems for Industrial IoT. Available online: <https://www.inpixon.com/> (accessed on 17 July 2024).
51. Eliko UWB RTLS—The Most Accurate and Reliable Tracking. Available online: <https://eliko.tech/uwb-rtls-ultra-wideband-real-time-location-system/> (accessed on 17 July 2024).
52. Employee Tracking System with RTLS | Litum. Available online: <https://litum.com/rtls-rfid-employee-tracking/> (accessed on 17 July 2024).
53. Product-Woxu Wireless. Available online: <https://uwb.woxuwireless.com/views/product/productTag.html> (accessed on 17 July 2024).
54. Real-Time Location System (RTLS) for Indoor Tracking | Sewio RTLS. Available online: <https://www.sewio.net/> (accessed on 17 July 2024).
55. KKM | IoT Devices ODM & OEM Solutions Manufacturer. Available online: <https://www.kkmcn.com> (accessed on 17 July 2024).
56. Ultra-Wideband Technology » Products » KINEXON. Available online: <https://kinexon.com/products/uwb-technology> (accessed on 17 July 2024).
57. Pozyx's UWB RTLS Hardware Tracks Anything Ultra Accurate. Available online: <https://www.pozyx.io/products/pozyx-uwb-rtls> (accessed on 17 July 2024).
58. Ultra Wideband (UWB) Technology | Zebra. Available online: <https://www.zebra.com/content/zebra1/gb/en/products/location-technologies/ultra-wideband.html> (accessed on 17 July 2024).
59. Admin Product. Tsingol. Available online: https://www.tsingol.com/en/product_ (accessed on 17 July 2024).
60. Manifold, S. Ubisense—Ultra-Wideband Real-Time Location System (RTLS). Available online: <https://ubisense.com/> (accessed on 17 July 2024).
61. Kathrein Solutions—Your Partner in RFID, AutoID, IoT, EVR & AVI. Available online: <https://www.kathrein-solutions.com/> (accessed on 17 July 2024).
62. Ubudu Hybrid Hardware | Track the Location of Assets. Available online: <https://ubudu.com/products/hardware> (accessed on 17 July 2024).
63. Available online: <https://tracktio.com/> (accessed on 17 July 2024).
64. UWB RTLS Solutions for Engineering & Commercial Vehicles Tracking. Available online: <https://leisureautosafety.com/products-leisure-auto/uwb-rtls/> (accessed on 17 July 2024).
65. UWB, UWB Kit, UWB Solutions, RTLS, NFC, Contactless, Mobile and MIFARE Solutions. Available online: <https://www.themobileknowledge.com/> (accessed on 18 July 2024).
66. BeSpoon's RTLS Solutions: Transforming Smart Industries—Bespoon 2024. Available online: <https://tracktio.com/uwb-rtls/> (accessed on 25 July 2024).
67. Shankar, P.M. Overview. In *Fading and Shadowing in Wireless Systems*; Shankar, P.M., Ed.; Springer International Publishing: Cham, Switzerland, 2017; pp. 1–7, ISBN 978-3-319-53198-4.
68. Agrawal, A.; Kshetrimayum, R.S. Analysis of UWB communication over IEEE 802.15.3a channel by superseding lognormal shadowing by Mixture of Gamma distributions. *AEU-Int. J. Electron. Commun.* **2015**, *69*, 1795–1799. [CrossRef]
69. Liu, A.; Lin, S.; Wang, J.; Kong, X. A Succinct Method for Non-Line-of-Sight Mitigation for Ultra-Wideband Indoor Positioning System. *Sensors* **2022**, *22*, 8247. [CrossRef]
70. Ni, Z.; Zhou, H.; Zhang, G.; Xu, W. UWB Non-Line-of-Sight Recognition and Suppression Methodology. In Proceedings of the 2023 15th International Conference on Communication Software and Networks (ICCSN), Shenyang, China, 21–23 July 2023; pp. 313–318.
71. Yang, J.; Feng, D.; Chu, Y.; Guo, C.; Zhuang, Y. An Efficient Cooperative Positioning Scheme in Non-Line-of-Sight Environments. In Proceedings of the 2022 IEEE 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN), Beijing, China, 5–7 September 2022; pp. 1–8.

72. He, J.; Zhao, W.; Liu, Y.; Han, F. Non-line-of-sight ultra-wideband signal recognition using DWT-CNN. In Proceedings of the Fourth International Conference on Sensors and Information Technology (ICSI 2024), Xiamen, China, 5–7 January 2024; SPIE; Volume 13107, pp. 633–639.
73. Qin, Z.; Meng, Z.; Li, Z.; Gao, N.; Zhang, Z.; Meng, Q.; Zhen, D. Compensating the NLoS Occlusion Errors of UWB for Pedestrian Localization With MIMU. *IEEE Sens. J.* **2023**, *23*, 12146–12158. [\[CrossRef\]](#)
74. Su, C.; Zhang, F.; Jin, B.; Ma, J.; Ni, J.; Li, S.; Wang, Z. UWBLoc: NLoS Mitigation for Practical UWB Localization. In Proceedings of the 2023 IEEE Smart World Congress (SWC), Portsmouth, UK, 28–31 August 2023; pp. 1–8.
75. Sang, C.L.; Steinhagen, B.; Homburg, J.D.; Adams, M.; Hesse, M.; Rückert, U. Identification of NLOS and Multi-Path Conditions in UWB Localization Using Machine Learning Methods. *Appl. Sci.* **2020**, *10*, 3980. [\[CrossRef\]](#)
76. Liu, Q.; Yin, Z.; Zhao, Y.; Wu, Z.; Wu, M. UWB LOS/NLOS identification in multiple indoor environments using deep learning methods. *Phys. Commun.* **2022**, *52*, 101695. [\[CrossRef\]](#)
77. Stahlke, M.; Kram, S.; Mutschler, C.; Mahr, T. NLOS Detection using UWB Channel Impulse Responses and Convolutional Neural Networks. In Proceedings of the 2020 International Conference on Localization and GNSS (ICL-GNSS), Tampere, Finland, 2–4 June 2020; pp. 1–6.
78. Hua, C.; Zhao, K.; Dong, D.; Zheng, Z.; Yu, C.; Zhang, Y.; Zhao, T. Multipath Map Method for TDOA Based Indoor Reverse Positioning System with Improved Chan-Taylor Algorithm. *Sensors* **2020**, *20*, 3223. [\[CrossRef\]](#) [\[PubMed\]](#)
79. He, C.; Tang, Z. An UWB Positioning Algorithm Based on Clock Compensation and Filtering. In Proceedings of the 2023 Cross Strait Radio Science and Wireless Technology Conference (CSRSWTC), Guilin, China, 10–13 November 2023; pp. 1–3.
80. Al-Okby, M.F.R.; Roddelkopf, T.; Burgdorf, S.-J.; Thurow, K. Multi-Tag UWB-based Indoor Positioning System for Objects Tracking. In Proceedings of the 2024 IEEE 22nd Jubilee International Symposium on Intelligent Systems and Informatics (SISY), Pula, Croatia, 19–21 September 2024; pp. 415–422.
81. Cheng, J.; Jin, J.; Huang, Y.; Yu, P. Design of Positioning System for Underground Personnel and Electromechanical Equipment Using WSN. In Proceedings of the 2020 IEEE International Conference on Applied Superconductivity and Electromagnetic Devices (ASEMD), Tianjin, China, 16–18 October 2020; pp. 1–2.
82. Minh, T.L.; Xuan, D.T. Applying Kalman filter to UWB positioning with DS-TWR method in LOS/NLOS scenarios. In Proceedings of the 2021 International Symposium on Electrical and Electronics Engineering (ISEE), Ho Chi Minh, Vietnam, 15–16 April 2021; pp. 95–99.
83. Qi, M.; Xue, B.; Wang, W. Calibration and Compensation of Anchor Positions for UWB Indoor Localization. *IEEE Sens. J.* **2024**, *24*, 689–699. [\[CrossRef\]](#)
84. Sidorenko, J.; Schatz, V.; Scherer-Negenborn, N.; Arens, M.; Hugentobler, U. Error Corrections for Ultrawideband Ranging. *IEEE Trans. Instrum. Meas.* **2020**, *69*, 9037–9047. [\[CrossRef\]](#)
85. Laadung, T.; Ulp, S.; Alam, M.M.; Moullec, Y.L. Novel Active-Passive Two-Way Ranging Protocols for UWB Positioning Systems. *IEEE Sens. J.* **2022**, *22*, 5223–5237. [\[CrossRef\]](#)
86. Van Herbruggen, B.; Van Leemput, D.; Van Landschoot, J.; De Poorter, E. Real-Time Anchor Node Selection for Two-Way-Ranging (TWR) Ultra-Wideband (UWB) Indoor Positioning Systems. *IEEE Sens. Lett.* **2024**, *8*, 6002404. [\[CrossRef\]](#)
87. Liu, Y.; Sun, R.; Liu, J.; Fan, Y.; Li, L.; Zhang, Q. Research on the Positioning Method of Autonomous Mobile Robot in Structure Space Based on UWB. In Proceedings of the 2019 International Conference on High Performance Big Data and Intelligent Systems (HPBD&IS), Shenzhen, China, 9–11 May 2019; pp. 278–282.
88. Tiemann, J.; Wietfeld, C. Scalability, Real-Time Capabilities, and Energy Efficiency in Ultra-Wideband Localization. *IEEE Trans. Ind. Inform.* **2019**, *15*, 6313–6321. [\[CrossRef\]](#)
89. Van Herbruggen, B.; Luchie, S.; Wilssens, R.; De Poorter, E. Single Anchor Localization by Combining UWB Angle-of-Arrival and Two-Way-Ranging: An Experimental Evaluation of the DW3000. In Proceedings of the 2024 International Conference on Localization and GNSS (ICL-GNSS), Antwerp, Belgium, 25–27 June 2024; pp. 1–7.
90. Zhou, Y.; Wang, Y.; Zhong, Z. Study of Mobile Robot positioning based on Ultra-Wideband System. In Proceedings of the 2022 IEEE 5th International Conference on Electronic Information and Communication Technology (ICEICT), Hefei, China, 21–23 August 2022; pp. 595–598.
91. Ali, R.; Liu, R.; Nayyar, A.; Qureshi, B.; Cao, Z. Tightly Coupling Fusion of UWB Ranging and IMU Pedestrian Dead Reckoning for Indoor Localization. *IEEE Access* **2021**, *9*, 164206–164222. [\[CrossRef\]](#)
92. Wang, X.; Zhang, J.; Wang, Y.; Gong, Q.; Wei, Z.; Zhang, X.; Yuan, X.; Yan, X.; Liu, Y.; Dong, L. Ultra-wideband High Precision Positioning System Based on EDS-TWR and EKF Algorithm. In Proceedings of the 2024 IEEE 14th International Conference on Electronics Information and Emergency Communication (ICEIEC), Beijing, China, 24–25 May 2024; pp. 1–6.
93. Molnár, M.; Luspai, T. Development of an UWB based Indoor Positioning System. In Proceedings of the 2020 28th Mediterranean Conference on Control and Automation (MED), Saint-Raphaël, France, 16–19 June 2020; pp. 820–825.
94. Zhang, H.; Lu, S.; Zhou, Z.; Wang, H.; Yan, B.; Si, X. Downhole robot positioning system based on UWB. In Proceedings of the 2022 China Automation Congress (CAC), Xiamen, China, 25–27 November 2022; pp. 2444–2449.
95. Wei, L.; Jian, W.; Lei, Y.; Ziyang, S. Optimization of Outdoor Long-Range Localization Algorithm Based on DS-TWR. In Proceedings of the 2023 5th International Conference on Intelligent Control, Measurement and Signal Processing (ICMSP), Chengdu, China, 19–21 May 2023; pp. 1113–1117.

96. Dai, P.; Wang, S.; Xu, T.; Li, M.; Gao, F.; Xing, J.; Yao, L. Efficient Localization Algorithm With UWB Ranging Error Correction Model Based on Genetic Algorithm-Ant Colony Optimization-Backpropagation Neural Network. *IEEE Sens. J.* **2023**, *23*, 29906–29918. [\[CrossRef\]](#)
97. Li, K.; Ni, W.; Wei, B.; Guizani, M. An Experimental Study of Two-way Ranging Optimization in UWB-based Simultaneous Localization and Wall-Mapping Systems. In Proceedings of the 2022 International Wireless Communications and Mobile Computing (IWCMC), Dubrovnik, Croatia, 30 May–3 June 2022; pp. 871–876.
98. Shyam, S.; Juliet, S.; Ezra, K. A UWB System Model for Patient Tracking and Monitoring for Smart Healthcare. In Proceedings of the 2022 6th International Conference on Devices, Circuits and Systems (ICDCS), Coimbatore, India, 21–22 April 2022; pp. 32–37.
99. Liu, Q.; Shi, M.; Hu, W.; Jiang, H.; Xu, Z.; Qiao, S. Design of high-precision wireless UWB indoor positioning system. In Proceedings of the 2022 Cross Strait Radio Science & Wireless Technology Conference (CSRSWTC), Haidian, China, 17–18 December 2022; pp. 1–3.
100. Xue, Y.; Su, W.; Wang, H.; Yang, D.; Jiang, Y. DeepTAL: Deep Learning for TDOA-Based Asynchronous Localization Security with Measurement Error and Missing Data. *IEEE Access* **2019**, *7*, 122492–122502. [\[CrossRef\]](#)
101. Cho, J.; Jeong, S.; Lee, B. A Study on Anchor Placement and 3D Positioning Algorithm for UWB Application in Small Sites. *KSCE J. Civ. Eng.* **2024**, *28*, 4575–4587. [\[CrossRef\]](#)
102. Cho, J.; Lee, B. Optimal layout of four anchors to improve accuracy of Ultra-Wide band based indoor positioning. *Expert Syst. Appl.* **2024**, *255*, 124808. [\[CrossRef\]](#)
103. Pătru, G.-C.; Flueratoru, L.; Vasilescu, I.; Niculescu, D.; Rosner, D. FlexTDOA: Robust and Scalable Time-Difference of Arrival Localization Using Ultra-Wideband Devices. *IEEE Access* **2023**, *11*, 28610–28627. [\[CrossRef\]](#)
104. Bottigliero, S.; Milanese, D.; Sacconi, M.; Maggiora, R. A Low-Cost Indoor Real-Time Locating System Based on TDOA Estimation of UWB Pulse Sequences. *IEEE Trans. Instrum. Meas.* **2021**, *70*, 5502211. [\[CrossRef\]](#)
105. Pérez-Solano, J.J.; Ezpeleta, S.; Claver, J.M. Indoor localization using time difference of arrival with UWB signals and unsynchronized devices. *Ad Hoc Netw.* **2020**, *99*, 102067. [\[CrossRef\]](#)
106. Chiasson, D.; Lin, Y.; Kok, M.; Shull, P.B. Asynchronous Hyperbolic UWB Source-Localization and Self-Localization for Indoor Tracking and Navigation. *IEEE Internet Things J.* **2023**, *10*, 11655–11668. [\[CrossRef\]](#)
107. Ma, Q.; Zhang, P.; Qian, G.; Feng, S. An Improved Two-Step Weighted Least Squares Aided UWB High-Precision Indoor Positioning System for Moving Targets. In Proceedings of the 2021 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS), Shenyang, China, 10–11 December 2021; pp. 1–7.
108. Leng, J.; Ma, G.; Zhu, J.; Ma, H. Improved TDOA Two-Stage UWB Localization Algorithm For Indoor Mobile Robot. In Proceedings of the 2021 IEEE International Conference on Recent Advances in Systems Science and Engineering (RASSE), Tainan, Taiwan, 7–10 November 2022; pp. 1–5.
109. Kim, H.J.; Xie, Y.; Yang, H.; Lee, C.; Song, T.L. An Efficient Indoor Target Tracking Algorithm Using TDOA Measurements with Applications to Ultra-Wideband Systems. *IEEE Access* **2019**, *7*, 91435–91445. [\[CrossRef\]](#)
110. Zhang, Y.; Duan, L. A phase-difference-of-arrival assisted ultra-wideband positioning method for elderly care. *Measurement* **2021**, *170*, 108689. [\[CrossRef\]](#)
111. Zhang, Y.; Duan, L. Toward Elderly Care: A Phase-Difference-of-Arrival Assisted Ultra-Wideband Positioning Method in Smart Home. *IEEE Access* **2020**, *8*, 139387–139395. [\[CrossRef\]](#)
112. Botler, L.; Gashi, M.; Diwold, K.; Römer, K. Poster Abstract: Machine Learning-based Models for Phase-Difference-of-Arrival Measurements Using Ultra-Wideband Transceivers. In Proceedings of the 2022 21st ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), Virtual Conference, 4–6 May 2022; pp. 517–518.
113. Ge, F.; Shen, Y. Single-Anchor Ultra-Wideband Localization System Using Wrapped PDoA. *IEEE Trans. Mob. Comput.* **2022**, *21*, 4609–4623. [\[CrossRef\]](#)
114. Petukhov, N.I.; Korogodin, I.V.; Serov, S.A.; Masalkova, N.V.; Malyshev, A.P.; Efremov, M. Analysis of Measurements of UWB PDoA Local Navigation System with Different Baselines. In Proceedings of the 2022 IEEE 23rd International Conference of Young Professionals in Electron Devices and Materials (EDM), Altai, Russia, 30 June–4 July 2022; pp. 223–227.
115. Krška, J.; Gentner, C.; Navrátil, V. Angle of Arrival Measurements with Ultra-wide Band Transceivers: Design and Evaluation. In Proceedings of the 2024 18th European Conference on Antennas and Propagation (EuCAP), Glasgow, UK, 17–22 March 2024; pp. 1–5.
116. Luo, J.; Zhang, J.; Yang, H.; Guan, Y. All-Directional DOA Estimation for Ultra-Wideband Regular Tetrahedral Array Using Wrapped PDoA. *Sensors* **2022**, *22*, 1532. [\[CrossRef\]](#) [\[PubMed\]](#)
117. Diagne, S.; Val, T.; Farota, A.K.; Diop, B.; Assogba, O. Performances Analysis of a System of Localization by Angle of Arrival UWB Radio. *Int. J. Commun. Netw. Syst. Sci.* **2020**, *13*, 15–27. [\[CrossRef\]](#)
118. Sippel, E.; Hehn, M.; Koegel, T.; Gröschel, P.; Hofmann, A.; Brückner, S.; Geiss, J.; Schober, R.; Vossiek, M. Quasi-Coherent Phase-Based Localization and Tracking of Incoherently Transmitting Radio Beacons. *IEEE Access* **2021**, *9*, 133229–133239. [\[CrossRef\]](#)
119. Zhang, K.; Shen, C.; Bao, M.; Vaniushkina, D.; Kumushai, K.; Zang, L. Research on Optimization Algorithm Based on PDoA. In Proceedings of the 2021 IEEE 21st International Conference on Communication Technology (ICCT), Tianjin, China, 13–16 October 2021; pp. 1427–1430.

120. Müller, M.; Röhrig, C. Angle of Arrival Estimation in the Presence of Phase Difference Ambiguities. In Proceedings of the 2022 IEEE 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN), Beijing, China, 5–7 September 2022; pp. 1–7.
121. Chen, S.; Yin, D.; Niu, Y. A Survey of Robot Swarms' Relative Localization Method. *Sensors* **2022**, *22*, 4424. [\[CrossRef\]](#)
122. Xu, X.; Liu, X.; Zhao, B.; Yang, B. An Extensible Positioning System for Locating Mobile Robots in Unfamiliar Environments. *Sensors* **2019**, *19*, 4025. [\[CrossRef\]](#)
123. Cano, J.; Ding, Y.; Pages, G.; Chaumette, E.; Le Ny, J. A Robust Kalman Filter Based Approach for Indoor Robot Positioning with Multi-Path Contaminated UWB Data. In Proceedings of the ICASSP 2023—2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Rhodes Island, Greece, 4–10 June 2023; pp. 1–5.
124. Ranjan, R.; Shin, D.; Jung, Y.; Kim, S.; Yun, J.-H.; Kim, C.-H.; Lee, S.; Kye, J. Comparative Analysis of Integrated Filtering Methods Using UWB Localization in Indoor Environment. *Sensors* **2024**, *24*, 1052. [\[CrossRef\]](#)
125. Tang, M.; Zhou, B.; Zhong, X.; Liu, X.; Li, Q. Enhanced indoor positioning through human-robot collaboration. *Urban Info* **2024**, *3*, 7. [\[CrossRef\]](#)
126. Alghuraid, A.; Almalki, A.; Alnaim, A.; Almubarak, A.; Alhuwaidi, S.; Hussain, A.A.; Alsubaie, M. Development of Ultra-Wide-Band Indoor Positioning System Designed to Control The Position of a Grid Confined Robotic Platform. In Proceedings of the 2021 4th International Symposium on Advanced Electrical and Communication Technologies (ISAECT), Alkhobar, Saudi Arabia, 6–8 December 2021; pp. 1–5.
127. Pichler-Scheder, M.; Pendl, K.; Schmidt, R.; Petruzella, M.; Mühlberger, G. 3D Performance Analysis of a UWB Positioning System in a Dynamic Robotic Scenario. In Proceedings of the 2019 16th Workshop on Positioning, Navigation and Communications (WPNC), Bremen, Germany, 23–24 October 2019; pp. 1–6.
128. Wang, Y.-X.; Chang, C.-L. ROS-base Multi-Sensor Fusion for Accuracy Positioning and SLAM System. In Proceedings of the 2020 International Symposium on Community-centric Systems (CcS), Hachioji, Japan, 23–26 September 2020; pp. 1–6.
129. Cano, J.; Le Ny, J. Ranging-Based Localizability Optimization for Mobile Robotic Networks. *IEEE Trans. Robot.* **2023**, *39*, 2842–2860. [\[CrossRef\]](#)
130. Bostanci, B.; Tekkok, S.; Soyunmez, E.; Oguz-Ekim, P.; Yeganli, F. The LiDAR and UWB based Source Localization and Initialization Algorithms for Autonomous Robotic Systems. In Proceedings of the 2019 11th International Conference on Electrical and Electronics Engineering (ELECO), Bursa, Turkey, 28–30 November 2019; pp. 900–904.
131. Cao, Z.; Liu, R.; Yuen, C.; Athukorala, A.; Kiat Ng, B.K.; Mathanraj, M.; Tan, U.-X. Relative Localization of Mobile Robots with Multiple Ultra-WideBand Ranging Measurements. In Proceedings of the 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Prague, Czech Republic, 27 September–1 October 2021; pp. 5857–5863.
132. Clark, L.; Andre, C.; Galante, J.; Krishnamachari, B.; Psounis, K. TEAM: Trilateration for Exploration and Mapping with Robotic Networks. In Proceedings of the 2021 18th International Conference on Ubiquitous Robots (UR), Gangneung, Republic of Korea, 12–14 July 2021; pp. 539–546.
133. Rezende, A.M.C.; Júnior, G.P.C.; Fernandes, R.; Miranda, V.R.F.; Azpúrua, H.; Pessin, G.; Freitas, G.M. Indoor Localization and Navigation Control Strategies for a Mobile Robot Designed to Inspect Confined Environments. In Proceedings of the 2020 IEEE 16th International Conference on Automation Science and Engineering (CASE), Hong Kong, China, 20–21 August 2020; pp. 1427–1433.
134. Wegner, T.E.; Gedschold, J.; Kropp, G.; Trabert, J.; Kmec, M.; Del Galdo, G. Descending Staircase Detection for Service Robots based on M-Sequence UWB Radar. In Proceedings of the 2021 18th European Radar Conference (EuRAD), London, UK, 5–7 April 2022; pp. 29–32.
135. Lima, J.; Pinto, A.F.; Ribeiro, F.; Pinto, M.; Pereira, A.I.; Pinto, V.H.; Costa, P. Data Fusion Using Ultra Wideband Time-of-Flight Positioning for Mobile Robot Applications. In Proceedings of the 2023 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC), Tomar, Portugal, 26–27 April 2023; pp. 251–256.
136. Kefferpütz, K.; McGuire, K. Error-State Unscented Kalman-Filter for UAV Indoor Navigation. In Proceedings of the 2022 25th International Conference on Information Fusion (FUSION), Linköping, Sweden, 4–7 July 2022; pp. 1–8.
137. Wang, W.; Marelli, D.; Fu, M. Multiple-Vehicle Localization Using Maximum Likelihood Kalman Filtering and Ultra-Wideband Signals. *IEEE Sens. J.* **2021**, *21*, 4949–4956. [\[CrossRef\]](#)
138. Zhang, K.; Chen, P.; Ma, T.; Gao, S. On-Demand Precise Tracking for Energy-Constrained UAVs in Underground Coal Mines. *IEEE Trans. Instrum. Meas.* **2022**, *71*, 5500814. [\[CrossRef\]](#)
139. Zhang, Y.; Qi, J.; Wu, C.; Liu, Y.; Pu, J. Indoor UAV Formation System Based on UWB Positioning. In Proceedings of the 2023 42nd Chinese Control Conference (CCC), Tianjin, China, 24–26 July 2023; pp. 8545–8550.
140. Lin, H.-Y.; Zhan, J.-R. GNSS-denied UAV indoor navigation with UWB incorporated visual inertial odometry. *Measurement* **2023**, *206*, 112256. [\[CrossRef\]](#)
141. Niculescu, V.; Palossi, D.; Magno, M.; Benini, L. Energy-Efficient, Precise UWB-Based 3-D Localization of Sensor Nodes with a Nano-UAV. *IEEE Internet Things J.* **2023**, *10*, 5760–5777. [\[CrossRef\]](#)
142. Lee, J.; Moon, J.; Kim, S. UWB-based Multiple UAV Control System for Indoor Ground Vehicle Tracking. In Proceedings of the 2021 IEEE VTS 17th Asia Pacific Wireless Communications Symposium (APWCS), Virtual Conference, 30–31 August 2021; pp. 1–5.

143. Xu, H.; Wang, L.; Zhang, Y.; Qiu, K.; Shen, S. Decentralized Visual-Inertial-UWB Fusion for Relative State Estimation of Aerial Swarm. In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France, 31 May–31 August 2020; pp. 8776–8782.
144. Queralta, J.P.; Martínez Almansa, C.; Schiano, F.; Floreano, D.; Westerlund, T. UWB-based System for UAV Localization in GNSS-Denied Environments: Characterization and Dataset. In Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Las Vegas, NV, USA, 25–29 October 2020; pp. 4521–4528.
145. Ochoa-de-Eribe-Landaberea, A.; Zamora-Cadenas, L.; Peñagaricano-Muñoz, O.; Velez, I. UWB and IMU-Based UAV's Assistance System for Autonomous Landing on a Platform. *Sensors* **2022**, *22*, 2347. [[CrossRef](#)] [[PubMed](#)]
146. Brunacci, V.; De Angelis, A. Fusion of UWB and Magnetic Ranging Systems for Robust Positioning. *IEEE Trans. Instrum. Meas.* **2024**, *73*, 7500712. [[CrossRef](#)]
147. Deliparaschos, K.M.; Loizou, S.G.; Zolotas, A.C. A Modular UAV Hardware Platform for Aerial Indoor Navigation Research and Development. In Proceedings of the 2024 International Conference on Unmanned Aircraft Systems (ICUAS), Chania, Greece, 4–7 June 2024; pp. 398–404.
148. Meng, Q.; Song, Y.; Li, S.; Zhuang, Y. Resilient tightly coupled INS/UWB integration method for indoor UAV navigation under challenging scenarios. *Def. Technol.* **2023**, *22*, 185–196. [[CrossRef](#)]
149. Huang, X.; Zhang, C.; Wei, Z.; Jiang, H. Research on adaptive positioning system: A simplified base station scheme based on UWB-PDOA technology. In Proceedings of the 2023 4th International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE), Guangzhou, China, 25–27 August 2023; pp. 295–301.
150. Zhang, F.; Hong, S.; Ding, Y.; Yang, S.-H. Ultrawideband-Based Real-Time Positioning With Cascaded Wireless Clock Synchronization Method. *IEEE Internet Things J.* **2024**, *11*, 16731–16745. [[CrossRef](#)]
151. Pham, N.H.H.; Nguyen, M.A.; Sun, C.-C. Indoor Positioning System using UWB and Kalman filter to increase the accuracy of the Localization System. In Proceedings of the 2022 IEEE International Conference on Consumer Electronics—Taiwan, Taipei, Taiwan, 6–8 July 2022; IEEE: Taipei, Taiwan, 2022; pp. 339–340.
152. Lu, H.; Wang, T.; Ge, F.; Shen, Y. A robust UWB array localization scheme through passive anchor assistance. *China Commun.* **2021**, *18*, 1–13. [[CrossRef](#)]
153. Chen, H.; Dhekne, A. Spoofing Evident and Spoofing Deterrent Localization Using Ultrawideband (UWB) Active–Passive Ranging. *IEEE J. Indoor Seamless Position. Navig.* **2024**, *2*, 12–24. [[CrossRef](#)]
154. Marti, S.; Diethelm, H.; Mathis, H.; Lang, H.-D. Aperture-Coupled Patch Antennas for UWB Indoor Localization Systems in the 4 and 6 GHz Bands. In Proceedings of the 2022 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI), Denver, CO, USA, 10–15 July 2022; pp. 950–951.
155. Phoojaroenchanachai, C.; Suwatthikul, C.; Maneerat, K.; Hormsup, K.; Chinda, K.; Wisadsud, S.; Demeechai, T.; Kovavisaruch, L.; Kaemarungsi, K. An Extendable Auto-Position for UWB-based Indoor Positioning Systems. In Proceedings of the 2021 25th International Computer Science and Engineering Conference (ICSEC), Chiang Rai, Thailand, 18–20 November 2021; pp. 334–339.
156. Vecchia, D.; Corbalán, P.; Istomin, T.; Picco, G.P. TALLA: Large-scale TDoA Localization with Ultra-wideband Radios. In Proceedings of the 2019 International Conference on Indoor Positioning and Indoor Navigation (IPIN), Pisa, Italy, 30 September–3 October 2019; pp. 1–8.
157. Sang, C.L.; Adams, M.; Korthals, T.; Hörmann, T.; Hesse, M.; Rückert, U. A Bidirectional Object Tracking and Navigation System using a True-Range Multilateration Method. In Proceedings of the 2019 International Conference on Indoor Positioning and Indoor Navigation (IPIN), Pisa, Italy, 30 September–3 October 2019; pp. 1–8.
158. Sung, S.; Kim, H.; Jung, J.-I. Accurate Indoor Positioning for UWB-Based Personal Devices Using Deep Learning. *IEEE Access* **2023**, *11*, 20095–20113. [[CrossRef](#)]
159. Zhu, Y.; Huang, L.; Liu, J.; Cao, Z.; Chen, J.; Mu, Z. A Real-Time UWB Location and Tracking System Based on TWR-TDOA Estimation and a Simplified MPGA Layout Optimization. *Mob. Inf. Syst.* **2022**, *2022*, 1194169. [[CrossRef](#)]
160. Van Herbruggen, B.; Fontaine, J.; Poorter, E.D. Anchor Pair Selection for Error Correction in Time Difference of Arrival (TDoA) Ultra Wideband (UWB) Positioning Systems. In Proceedings of the 2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN), Lloret de Mar, Spain, 29 November–2 December 2021; pp. 1–8.
161. Wang, W.; Wang, G.; Ho, K.C.; Huang, L. Robust TDOA localization based on maximum correntropy criterion with variable center. *Signal Process.* **2023**, *205*, 108860. [[CrossRef](#)]
162. Lu, Y.-M.; Sheu, J.-P.; Kuo, Y.-C. Deep Learning for Ultra-Wideband Indoor Positioning. In Proceedings of the 2021 IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Helsinki, Finland, 13–16 September 2021; pp. 1260–1266.
163. Zhou, X.; Xu, C.; He, J.; Wan, J. A Cross-region Wireless-synchronization—based TDOA Method for Indoor Positioning Applications. In Proceedings of the 2019 28th Wireless and Optical Communications Conference (WOCC), Beijing, China, 9–10 May 2019; pp. 1–4.
164. Kolakowski, J. A Method for TDOA Errors Mitigation in UWB Positioning System. In Proceedings of the 2020 23rd International Microwave and Radar Conference (MIKON), Warsaw, Poland, 5–7 October 2020; pp. 148–151.
165. Cao, H.; Liu, J.; Zhu, C.; Jing, L.; Chen, T.; Wang, T. A Novel Data Fusion Method For Indoor Navigation Using IMU Aided UWB. In Proceedings of the 2019 IEEE 2nd International Conference on Electronic Information and Communication Technology (ICEICT), Harbin, China, 20–22 January 2019; pp. 100–105.

166. Martalò, M.; Perri, S.; Verdano, G.; De Mola, F.; Monica, F.; Ferrari, G. Improved UWB TDoA-Based Positioning Using a Single Hotspot for Industrial IoT Applications. *IEEE Trans. Ind. Inform.* **2022**, *18*, 3915–3925. [\[CrossRef\]](#)
167. Schmickmann, F.; Haferkamp, M.; Tiemann, J.; Wietfeld, C. Bring Your Own Positioning System: An Infrastructure-free and Omnidirectional UWB-based Localization Approach. In Proceedings of the 2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring), Florence, Italy, 20–23 June 2023; pp. 1–7.
168. Divya, B.; Veeramakali, T.; Revathi, S.; Harisharajan, R.R.; Adhirooban, D.; Ebishdon, G.V. Indoor Navigation Using Ultra Wide Band. In Proceedings of the 2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS), Chennai, India, 14–15 December 2023; pp. 1–9.
169. Bhattacharya, S.; Choi, J. RNN-Based Robust Smartphone Indoor Localization on Ultra-Wideband DL-TDOA. In Proceedings of the 2023 IEEE International Conference on Communications Workshops (ICC Workshops), Rome, Italy, 28 May–1 June 2023; pp. 500–505.
170. Tian, D.; Xiang, Q. Research on Indoor Positioning System Based on UWB Technology. In Proceedings of the 2020 IEEE 5th Information Technology and Mechatronics Engineering Conference (ITOEC), Chongqing, China, 12–14 June 2020; pp. 662–665.
171. Lou, X.; Zhao, Y. High-Accuracy Positioning Algorithm Based on UWB. In Proceedings of the 2019 International Conference on Artificial Intelligence and Advanced Manufacturing (AIAM), Dublin, Ireland, 17–19 October 2019; pp. 71–75.
172. Malyshev, A.; Chugunov, A.; Chernyh, S. Investigation of a Novel Architecture for a One-Way Ranging UWB Local Navigation System. In Proceedings of the 2024 International Russian Smart Industry Conference (SmartIndustryCon), Sochi, Russia, 24–30 March 2024; pp. 333–338.
173. Pudlovskiy, V.; Chugunov, A.; Kulikov, R. Investigation of Impact of UWB RTLS Errors on AGV Positioning Accuracy. In Proceedings of the 2019 International Russian Automation Conference (RusAutoCon), Sochi, Russia, 8–14 September 2019; pp. 1–5.
174. Blake, K.; Huey, R.; Menezes, D.; Root, J.; Tran, L.; Chen, H. Robot Navigation Using Ultra-Wideband Indoor Localization and Dead Reckoning Algorithms. In Proceedings of the 2022 Opportunity Research Scholars Symposium (ORSS), Atlanta, GA, USA, 27 April 2022; pp. 12–15.
175. Huang, L.; Chen, R.; Chen, Y.; Guo, G.; Ye, F.; Liu, Z.; Chen, L. Robust TDOA-Based Indoor Localization Using Improved Clock-Sync-Scheme and Multilevel Constrained ARPF. *IEEE Sens. J.* **2023**, *23*, 10633–10643. [\[CrossRef\]](#)
176. Shyam, S.; Juliet, S.; Ezra, K. Vehicle Monitoring and Localization using UWB in Complex Parking Lot. In Proceedings of the 2022 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS), Erode, India, 7–9 April 2022; pp. 937–943.
177. Chen, H.; Dhekne, A. UnSpoof: Distance Spoofing-Evident Localization using UWB. In Proceedings of the 2023 13th International Conference on Indoor Positioning and Indoor Navigation (IPIN), Nuremberg, Germany, 25–28 September 2023; pp. 1–6.
178. Zhang, F.; Yang, L.; Liu, Y.; Ding, Y.; Yang, S.-H.; Li, H. Design and Implementation of Real-Time Localization System (RTLS) Based on UWB and TDoA Algorithm. *Sensors* **2022**, *22*, 4353. [\[CrossRef\]](#)
179. Grasso, P.; Innocente, M.S.; Tai, J.J.; Haas, O.; Dizqah, A.M. Analysis and Accuracy Improvement of UWB-TDoA-Based Indoor Positioning System. *Sensors* **2022**, *22*, 9136. [\[CrossRef\]](#)
180. Krška, J.; Navrátil, V. Utilization of Carrier-Frequency Offset Measurements in UWB TDoA Positioning with Receiving Tag. *Sensors* **2023**, *23*, 2595. [\[CrossRef\]](#)
181. Djosic, S.; Stojanovic, I.; Jovanovic, M.; Djordjevic, G.L. Multi-algorithm UWB-based localization method for mixed LOS/NLOS environments. *Comput. Commun.* **2022**, *181*, 365–373. [\[CrossRef\]](#)
182. Waqar, A.; Ahmad, I.; Habibi, D.; Phung, Q.V. Analysis of GPS and UWB positioning system for athlete tracking. *Meas. Sens.* **2021**, *14*, 100036. [\[CrossRef\]](#)
183. Hindermann, P.; Nüesch, S.; Früh, D.; Rüst, A.; Gyax, L. High precision real-time location estimates in a real-life barn environment using a commercial ultra wideband chip. *Comput. Electron. Agric.* **2020**, *170*, 105250. [\[CrossRef\]](#)
184. Wang, Y.; Lu, Y.; Zhou, Y.; Zhao, Z. Maximum Correntropy Criterion-Based UKF for Loosely Coupling INS and UWB in Indoor Localization. *CMES* **2024**, *139*, 2673–2703. [\[CrossRef\]](#)
185. Sagar, A.S.; Kim, T.; Park, S.; Lee, H.; Kim, H. Relative-Position Estimation Based on Loosely Coupled UWB-IMU Fusion for Wearable IoT Devices. *CMC* **2023**, *75*, 1941–1961. [\[CrossRef\]](#)
186. Zhou, H.; Yao, Z.; Zhang, Z.; Liu, P.; Lu, M. An Online Multi-Robot SLAM System Based on Lidar/UWB Fusion. *IEEE Sens. J.* **2022**, *22*, 2530–2542. [\[CrossRef\]](#)
187. Zayets, A.; Gentner, C.; Steinbach, E. High-Precision Multipath-Based Indoor Localization Scheme With User Privacy Protection for Dynamic NLoS Environments. *IEEE Access* **2021**, *9*, 116033–116049. [\[CrossRef\]](#)
188. Chen, Z.; Xu, A.; Sui, X.; Wang, C.; Wang, S.; Gao, J.; Shi, Z. Improved-UWB/LiDAR-SLAM Tightly Coupled Positioning System with NLOS Identification Using a LiDAR Point Cloud in GNSS-Denied Environments. *Remote Sens.* **2022**, *14*, 1380. [\[CrossRef\]](#)
189. Yang, H.; Kuang, Y.; Wang, M.; Bao, X.; Yang, Y. Indoor Mobile Localization Based on A Tightly Coupled UWB/INS Integration. In Proceedings of the 2020 16th International Conference on Control, Automation, Robotics and Vision (ICARCV), Shenzhen, China, 13–15 December 2020; pp. 1354–1359.
190. Zhao, W.; Panerati, J.; Schoellig, A.P. Learning-Based Bias Correction for Time Difference of Arrival Ultra-Wideband Localization of Resource-Constrained Mobile Robots. *IEEE Robot. Autom. Lett.* **2021**, *6*, 3639–3646. [\[CrossRef\]](#)

191. Tu, C.; Li, J.; Fu, D.; Wang, X. Multi-robot Positioning and Anti-interference Based on Ultra Wide Band. In *Progressive and Integrative Ideas and Applications of Engineering Systems Under the Framework of IOT and AI, Proceedings of the 2nd International Conference on Intelligent Systems Design and Engineering Applications, Okayama, Japan, 12–14 May 2023*; Ma, Y., Ed.; Springer Nature: Singapore, 2024; pp. 43–51.
192. Slovák, J.; Kurčová, P.; Vachálek, J.; Fit'ka, I. Navigation of the mobile robots using the RTLS technology implemented in the ROS. In *Proceedings of the 2020 Cybernetics & Informatics (K&I), Velke Karlovice, Czech Republic, 29 January–1 February 2020*; pp. 1–5.
193. Chengbo, K.; Huibin, Y.; Juan, Y.; Wenchao, L. Research on Wheeled Mobile Robot Positioning Based on EKF Multi-sensor Data Fusion. In *Proceedings of the 2019 3rd International Conference on Electronic Information Technology and Computer Engineering (EITCE), Xiamen, China, 18–20 October 2019*; pp. 457–460.
194. Zhang, H.; Zhang, Z.; Gao, N.; Xiao, Y.; Meng, Z.; Li, Z. Cost-Effective Wearable Indoor Localization and Motion Analysis via the Integration of UWB and IMU. *Sensors* **2020**, *20*, 344. [\[CrossRef\]](#)
195. Cao, Y.; Yang, C.; Li, R.; Knoll, A.; Beltrame, G. Accurate position tracking with a single UWB anchor. In *Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France, May 31–August 31 2020*; pp. 2344–2350.
196. Liu, F.; Zhang, J.; Wang, J.; Han, H.; Yang, D. An UWB/Vision Fusion Scheme for Determining Pedestrians' Indoor Location. *Sensors* **2020**, *20*, 1139. [\[CrossRef\]](#)
197. Wang, Y.; Zhang, D.; Li, Z.; Lu, M.; Zheng, Y.; Fang, T. System and Method for Reducing NLOS Errors in UWB Indoor Positioning. *Appl. Sci.* **2024**, *14*, 5123. [\[CrossRef\]](#)
198. Gao, Y.; Ma, W.; Cao, J.; Qu, J. Range-Only UWB SLAM for Indoor Robot Localization Employing Multi-Interval EFIR Rauch-Tung-Striebel Smoother. *CMES* **2021**, *130*, 1221–1237. [\[CrossRef\]](#)
199. Kolakowski, M.; Djaja-Josko, V.; Kolakowski, J. Static LiDAR Assisted UWB Anchor Nodes Localization. *IEEE Sens. J.* **2022**, *22*, 5344–5351. [\[CrossRef\]](#)
200. Marković, L.; Kovač, M.; Milijas, R.; Car, M.; Bogdan, S. Error State Extended Kalman Filter Multi-Sensor Fusion for Unmanned Aerial Vehicle Localization in GPS and Magnetometer Denied Indoor Environments. In *Proceedings of the 2022 International Conference on Unmanned Aircraft Systems (ICUAS), Dubrovnik, Croatia, 21–24 June 2022*; pp. 184–190.
201. Santoro, L.; Brunelli, D.; Fontanelli, D. On-Line Optimal Ranging Sensor Deployment for Robotic Exploration. *IEEE Sens. J.* **2022**, *22*, 5417–5426. [\[CrossRef\]](#)
202. Chen, S.; Xing, A.; Li, X.; Wang, Y.; Liu, W. Robust UWB Navigation System for UAV Swarm. In *Proceedings of the 2023 9th International Conference on Mechatronics and Robotics Engineering (ICMRE), Shenzhen, China, 10–12 February 2023*; pp. 101–105.
203. Zhan, J.-R.; Lin, H.-Y. Improving Visual Inertial Odometry with UWB Positioning for UAV Indoor Navigation. In *Proceedings of the 2022 26th International Conference on Pattern Recognition (ICPR), Montreal, QC, Canada, 21–25 August 2022*; pp. 4189–4195.
204. Pourjabar, M.; AlKatheeri, A.; Rusci, M.; Barcis, A.; Niculescu, V.; Ferrante, E.; Palossi, D.; Benini, L. Land & Localize: An Infrastructure-free and Scalable Nano-Drones Swarm with UWB-based Localization. In *Proceedings of the 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Pafos, Cyprus, 19–21 June 2023*; pp. 654–660.
205. Jia, J.; Guo, K.; Li, W.; Yu, X.; Guo, L. Composite Filtering for UWB-Based Localization of Quadrotor UAV With Skewed Measurements and Uncertain Dynamics. *IEEE Trans. Instrum. Meas.* **2022**, *71*, 1002313. [\[CrossRef\]](#)
206. Zhang, Q.; Wei, Y.; Li, X.; Xu, H. The positioning system use for autonomous navigation of unmanned aerial vehicles. In *Proceedings of the 2021 33rd Chinese Control and Decision Conference (CCDC), Kunming, China, 22–24 May 2021*; pp. 5064–5069.
207. Cao, Y.; Beltrame, G. VIR-SLAM: Visual, inertial, and ranging SLAM for single and multi-robot systems. *Auton. Robot.* **2021**, *45*, 905–917. [\[CrossRef\]](#)
208. Kao, P.-Y.; Chang, H.-J.; Tseng, K.-W.; Chen, T.; Luo, H.-L.; Hung, Y.-P. VIUNet: Deep Visual-Inertial-UWB Fusion for Indoor UAV Localization. *IEEE Access* **2023**, *11*, 61525–61534. [\[CrossRef\]](#)
209. Yang, B.; Yang, E.; Yu, L.; Loeliger, A. High-Precision UWB-Based Localisation for UAV in Extremely Confined Environments. *IEEE Sens. J.* **2022**, *22*, 1020–1029. [\[CrossRef\]](#)
210. Santoro, L.; Nardello, M.; Brunelli, D.; Fontanelli, D. UWB-Based Indoor Positioning System With Infinite Scalability. *IEEE Trans. Instrum. Meas.* **2023**, *72*, 1005711. [\[CrossRef\]](#)
211. Chen, C.; Kia, S.S. OptiTrack-Aided Supervised Learning for Neural Network-Based Ultra-Wideband Ranging Bias Correction. *IEEE Sens. J.* **2024**, *24*, 8484–8492. [\[CrossRef\]](#)
212. Morón, P.T.; Salimpour, S.; Fu, L.; Yu, X.; Queralta, J.P.; Westerlund, T. Benchmarking UWB-Based Infrastructure-Free Positioning and Multi-Robot Relative Localization: Dataset and Characterization. In *Proceedings of the 2023 IEEE Sensors Applications Symposium (SAS), Ottawa, ON, Canada, 18–20 July 2023*; pp. 1–6.
213. Chen, H.; Dhekne, A. PnPLoc: UWB Based Plug & Play Indoor Localization. In *Proceedings of the 2022 IEEE 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN), Beijing, China, 5–7 September 2022*; pp. 1–8.
214. Santoro, L.; Nardello, M.; Brunelli, D.; Fontanelli, D. Scale up to infinity: The UWB Indoor Global Positioning System. In *Proceedings of the 2021 IEEE International Symposium on Robotic and Sensors Environments (ROSE), Virtual Conference, 28–29 October 2021*; pp. 1–8.

215. Mayer, P.; Magno, M.; Benini, L. Self-Sustaining Ultrawideband Positioning System for Event-Driven Indoor Localization. *IEEE Internet Things J.* **2024**, *11*, 1272–1284. [\[CrossRef\]](#)
216. Plangger, J.; Govindasamy Ravichandran, H.; Rodin, S.C.; Atia, M. System Design and Performance Analysis of Indoor Real-time Localization using UWB Infrastructure. In Proceedings of the 2023 IEEE International Systems Conference (SysCon), Vancouver, BC, Canada, 17–20 April 2023; pp. 1–8.
217. Kim, D.-H.; Farhad, A.; Pyun, J.-Y. UWB Positioning System Based on LSTM Classification With Mitigated NLOS Effects. *IEEE Internet Things J.* **2023**, *10*, 1822–1835. [\[CrossRef\]](#)
218. Barbieri, L.; Brambilla, M.; Pitic, R.; Trabattini, A.; Mervic, S.; Nicoli, M. UWB Real-Time Location Systems for Smart Factory: Augmentation Methods and Experiments. In Proceedings of the 2020 IEEE 31st Annual International Symposium on Personal, Indoor and Mobile Radio Communications, London, UK, 31 August–3 September 2020; pp. 1–7.
219. Adame, T.; Igual, J.; Catalan, M. Fast Deployment of a UWB-Based IPS for Emergency Response Operations. *Sensors* **2023**, *23*, 4193. [\[CrossRef\]](#) [\[PubMed\]](#)
220. Hu, X.; Assaad, R.H. A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies. *J. Build. Eng.* **2024**, *86*, 108901. [\[CrossRef\]](#)
221. Niu, Z.; Yang, H.; Zhou, L.; Farag Taha, M.; He, Y.; Qiu, Z. Deep learning-based ranging error mitigation method for UWB localization system in greenhouse. *Comput. Electron. Agric.* **2023**, *205*, 107573. [\[CrossRef\]](#)
222. Song, Y.; Hsu, L.-T. Tightly coupled integrated navigation system via factor graph for UAV indoor localization. *Aerosp. Sci. Technol.* **2021**, *108*, 106370. [\[CrossRef\]](#)
223. Cheng, L.; Zhao, Z.; Shi, Y.; Lu, Y. Implicit unscented particle filter based indoor fusion positioning algorithms for sensor networks. *Alex. Eng. J.* **2024**, *94*, 104–119. [\[CrossRef\]](#)
224. Wei, J.; Wang, H.; Su, S.; Tang, Y.; Guo, X.; Sun, X. NLOS identification using parallel deep learning model and time-frequency information in UWB-based positioning system. *Measurement* **2022**, *195*, 111191. [\[CrossRef\]](#)
225. Guan, M.; Wen, C. Autonomous exploration using UWB and LiDAR. *J. Autom. Intell.* **2023**, *2*, 51–60. [\[CrossRef\]](#)
226. Djosic, S.; Stojanovic, I.; Jovanovic, M.; Nikolic, T.; Djordjevic, G.L. Fingerprinting-assisted UWB-based localization technique for complex indoor environments. *Expert Syst. Appl.* **2021**, *167*, 114188. [\[CrossRef\]](#)
227. Zang, L.; Jiao, J.; Wang, J.; Ding, N.; Wang, Z.; Min, X. Angle-Weighted trilateration method fusing multi-data processing technologies in indoor scene. *Digit. Signal Process.* **2024**, *147*, 104417. [\[CrossRef\]](#)
228. Li, C.-H.; Lai, A.I.-C.; Wu, R.-B. Accuracy enhancement of Wi-Fi positioning by ranging in dense UWB networks. *Comput. Commun.* **2024**, *213*, 179–187. [\[CrossRef\]](#)
229. Qian, L.; Chan, A.; Cai, J.; Lewicke, J.; Gregson, G.; Lipsett, M.; Rios Rincón, A. Evaluation of the accuracy of a UWB tracker for in-home positioning for older adults. *Med. Eng. Phys.* **2024**, *126*, 104155. [\[CrossRef\]](#) [\[PubMed\]](#)
230. Shyam, S.; Devaraj, S.J.; Ezra, K.; Delattre, J.; Lynus, G.K. Chapter 10—Design and implementation of UWB-based cyber-physical system for indoor localization in an industry environment. In *Intelligent Edge Computing for Cyber Physical Applications*; Hemanth, D.J., Gupta, B.B., Elhoseny, M., Shinde, S.V., Eds.; Intelligent Data-Centric Systems; Academic Press: New York, NY, USA, 2023; pp. 167–185, ISBN 978-0-323-99412-5.
231. Morawska, B.; Lipinski, P.; Lichy, K.; Adamkiewicz, K. Transfer learning-based UWB indoor localization using MHT-MDC and clusterization-based sparse fingerprinting. *J. Comput. Sci.* **2022**, *61*, 101654. [\[CrossRef\]](#)
232. Bregar, K. Indoor UWB Positioning and Position Tracking Data Set. *Sci. Data* **2023**, *10*, 744. [\[CrossRef\]](#) [\[PubMed\]](#)
233. Kia, G.; Ruotsalainen, L.; Talvitie, J. Toward Accurate Indoor Positioning: An RSS-Based Fusion of UWB and Machine-Learning-Enhanced WiFi. *Sensors* **2022**, *22*, 3204. [\[CrossRef\]](#)
234. Lou, P.; Zhao, Q.; Zhang, X.; Li, D.; Hu, J. Indoor Positioning System with UWB Based on a Digital Twin. *Sensors* **2022**, *22*, 5936. [\[CrossRef\]](#)
235. Zeng, Y.; Shen, S. Eliminating UWB Self Positioning Error Based on Kalman Filter Algorithm. In Proceedings of the Proceedings of the 2022 2nd International Conference on Bioinformatics and Intelligent Computing, Harbin, China, 21–23 January 2022; Association for Computing Machinery: New York, NY, USA, 2022; pp. 217–222.
236. Zhang, B.; Shen, L.; Yao, J.; Tang, S.-K.; Mirri, S. UWB Hybrid Filtering-Based Mobile IoT Device Tracking. In Proceedings of the Proceedings of the 2023 ACM Conference on Information Technology for Social Good, Lisbon, Portugal, 6–8 September 2023; Association for Computing Machinery: New York, NY, USA, 2023; pp. 471–476.
237. Zhang, B.; Zhan, H.; Hou, Y.; Chen, X. Ultra-wideband/BDS Indoor and Outdoor Seamless Positioning Algorithm Based on FNN-WCF. In Proceedings of the Proceedings of the 2022 5th International Conference on Artificial Intelligence and Pattern Recognition, Chengdu, China, 19–21 August 2022; Association for Computing Machinery: New York, NY, USA, 2023; pp. 1120–1125.
238. Neuhold, D.; Fakhreddine, A.; Bettstetter, C. HiPR+: A Protocol for Centimeter 3D Localization based on UWB. In Proceedings of the Proceedings of the 25th International ACM Conference on Modeling Analysis and Simulation of Wireless and Mobile Systems, Montreal, QC, Canada, 24–28 October 2022; Association for Computing Machinery: New York, NY, USA, 2022; pp. 181–188.
239. Zhao, J.; Hou, Q.; Xu, Z.; Lu, D.; Yang, G.; Shi, H. Design and Implementation of Indoor Positioning System Based on STM32F103C8T6. In Proceedings of the International Conference on Frontiers of Electronics, Information and Computation Technologies, Changsha, China, 21–23 May 2021; Association for Computing Machinery: New York, NY, USA, 2022; pp. 1–6.

240. Großwindhager, B.; Stocker, M.; Rath, M.; Boano, C.A.; Römer, K. SnapLoc: An ultra-fast UWB-based indoor localization system for an unlimited number of tags. In Proceedings of the Proceedings of the 18th International Conference on Information Processing in Sensor Networks, Montreal, QC, Canada, 15–18 April 2019; Association for Computing Machinery: New York, NY, USA, 2019; pp. 61–72.
241. Yang, J.; Dong, B.; Wang, J. VULoc: Accurate UWB Localization for Countless Targets without Synchronization. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* **2022**, *6*, 1–25. [[CrossRef](#)]
242. Zhao, M.; Chang, T.; Arun, A.; Ayyalasomayajula, R.; Zhang, C.; Bharadia, D. ULoc: Low-Power, Scalable and cm-Accurate UWB-Tag Localization and Tracking for Indoor Applications. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*; Association for Computing Machinery: New York, NY, USA, 2021; Volume 5, pp. 1–31. [[CrossRef](#)]
243. Polonelli, T.; Schläpfer, S.; Magno, M. Performance Comparison between Decawave DW1000 and DW3000 in low-power double side ranging applications. In Proceedings of the 2022 IEEE Sensors Applications Symposium (SAS), Sundsvall, Sweden, 1–3 August 2022; pp. 1–6.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.